

An active decentralized algorithm to maintain space swarm

Investigation on a satellite swarm control algorithm

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Declaration

I affirm that the investigation and results presented in this thesis is my original work and has not been previously submitted, in whole or part, for any degree or diploma at this institution or any other university

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Abstract

This study investigates the challenge of maintaining a space swarm for a group of model satellites. Each satellite is assigned random initial velocities and positions, and the control algorithms employed are based on artificial potentials. Attractive potentials are placed on the goal, while repulsive potentials are assigned to obstacles. The primary objective is to ensure the viability of these control potentials, prevent divergence of the group of model satellites under control, and achieve a stable formation without centralized control. Two main approaches are examined for their effectiveness in controlling the movement and location of the satellite models. In the first approach, the focus is on maintaining a stable formation using artificial potentials, and checking the viability on the proposed control algorithm. The artificial potentials consist of both attractive and repulsive potentials. The attractive potentials guide the model satellites towards a specified goal location, while the repulsive potentials prevent collisions with obstacles and each other. This decentralized method aims to ensure that the satellites move cohesively as a swarm without requiring a central command system. The second approach further refines the use of artificial potentials. Rather than letting the system determine its formation, a given formation shape is given. This involves fine-tuning the parameters governing the attractive and repulsive forces to achieve this formation. Both approaches are evaluated through numerical analysis and visual simulations. Numerically, defined

parameters allow for fine control over the system, providing a quantitative measure of the satellites' performance. Visually, real-time simulations offer a qualitative assessment of each model's motion and interaction within the swarm. A custom 3D simulation environment is developed using C++, OpenGL, with GLEW and GLFW. This environment allows for comprehensive control over all variables in the study, facilitating detailed analysis and adjustment of the swarm's behavior. By integrating these technologies, the simulation provides an interactive and flexible platform for testing and refining the control algorithms. With advanced simulation technologies, an interactive platform for testing and optimizing control algorithms for satellite swarm management is crucial for studying and testing algorithms. This study contributes to advancing decentralized control methodologies for satellite missions, crucial for applications in Earth observation, environmental monitoring, debris removal, and scientific research. Future research may focus on enhancing the algorithm robustness, exploring additional formation shapes, and optimizing control parameters to further improve swarm performance and efficiency in dynamic space environments.

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1. Introduction

Artificial potential fields have been widely utilized in robotics and autonomous systems, providing a versatile framework for path planning, Collision avoidance and formation control. This study explores the application of Artificial potential fields in the context of satellite swarms to control multiple satellites operating in coordination to achieve various mission objectives. The decentralized nature Artificial potential fields offer advantages over centralized control systems partially in environments characterized by communication delays, and need to autonomous decision making. The use of Artificial potential in robotics dates back several decades and have been instrumental in advancing autonomous navigation and control. Gazi and Passino (2003) from Ohio State University investigated the use of artificial potentials for robot swarm control [\[7\]](#). Their work showed that artificial potential fields could guide multiple robots to a goal while ensuring collision avoidance through repulsive potentials. In 2011, Roi and Choi from South Korea advanced instituted of science and technology (KAIST) implemented artificial potentials for satellite formation flying, focusing on maintaining relative positions in a predetermined formation [\[8\]](#). They demonstrated the effectiveness of potential fields in achieving and maintaining a satellite formation, but pointed out the problem of local minima (here satellites could become stuck in non-optimal positions). Beard et al. (2001) from Brigham Young University developed decentralized control algorithms for spacecraft

formation flying, incorporating potential [\[9\]](#). Their research underscored the challenges of balancing attractive and repulsive forces to maintain formation stability and avoid divergence. Patterson and Kingston (2002) from the Air Force Research Laboratory who worked on control of satellite constellations using artificial potential [\[10\]](#), emphasized the need for each satellite to make autonomous decisions based on local information. With advancement in space technology, this has led to a growing interest in satellite swarms, which are multiple satellites operating in coordination in order to achieve a variety of objectives. A decentralized approach offers numerous advantages over centralized control algorithms, such as enhanced coverage, redundancy, and flexibility. Satellite swarms are particularly beneficial for applications such as Earth observation, environmental monitoring, debris removal, scientific research, and communication networks. However, managing a swarm of satellites poses significant challenges:

- Maintaining a cohesive formation while navigating is essential. Artificial potentials can be tailored to provide attractive forces to maintain formation integrity and repulsive forces to prevent collisions.
- Satellites should autonomously detect and avoid collisions with space debris, other satellites, or natural celestial bodies. Artificial potentials can dynamically adjust repulsive potentials based on proximity and relative velocities to ensure safe navigation.

- Satellite swarms should be flexible and be able to adapt to changing mission requirements and operational conditions. Artificial potentials allow for real-time adjustments in formation shape, density, and trajectory based on updated mission objectives or unexpected environmental changes.
- Optimizing fuel consumption is critical for extending mission duration and reducing operational costs. Artificial potentials can assist in planning efficient trajectories and minimizing unnecessary maneuvers through proactive path planning strategies.

There is a need for a decentralized control systems that ensure stable formations and effective navigation without relying on a central command unit. One of the fundamental challenges in maintaining a satellite swarm is ensuring that each satellite attaches cohesively in a formation while navigating through space. Given the vast and dynamic nature of the space environment, satellites must continuously adjust their positions and velocities to avoid collisions with obstacles and each other, all while pursuing specific goals such as reaching designated orbits or covering particular areas of interest. Traditional control methods, which often rely on centralized systems, are not feasible for satellite swarms due to communication delays, potential failures, and the high degree of autonomy required for each satellite. In addressing these challenges, this dissertation investigates the use of artificial potentials as a control mechanism for satellite swarms. Artificial potential

fields are a well-established concept in robotics and autonomous systems, where virtual forces are used to guide the movement of agents. In this context, attractive potentials are applied to draw satellites towards target positions or formations, while repulsive potentials prevent collisions by pushing satellites away from obstacles and each other. This decentralized approach enables each satellite to operate autonomously, adjusting its trajectory based on local interactions and the overall swarm behavior.

2. Dynamic equations of motion and Control

2.1. Relative translational equations of motion

Consider a varied number of satellites moving on an orbit. The satellites initially are at a certain distance from each other. They should be able to keep a formation as they move of this orbit. The main study of this dissertation is to maintain this formation and prevent divergence. It is assumed that the satellite is equipped with onboard propulsion system able to provide continuous thrust. The motion of or model satellite is described by provide a linearized model for the relative translational motion of two nearby orbiting objects. One called the chief which moves on an orbit, and the chaser is called deputy. These relative translational equations of motion are called Clohessy-Wiltshire Equations and are as follow:

$$\ddot{x} = 3n^2x + 2n\dot{y} \quad (1.1)$$

$$\ddot{y} = -2n\dot{x} \quad (1.2)$$

$$\ddot{z} = -n^2z \quad (1.3)$$

Where, $n = \sqrt{\mu/R^3}$ is the orbital rate, μ = standard gravitational parameter, R is the radius of the target body's circular orbit,

These equations were gotten from [6] . from [4],Equations (1.1) – (1.3) are the Clohessy-Wiltshire equations with no disturbing and/or control accelerations. The nonhomogeneous forms of equations. (1.1) – (1.3) are:

$$\ddot{x} = 3n^2x + 2n\dot{y} + d_x + a_x \quad (1.4)$$

$$\ddot{y} = -2n\dot{x} + d_y + a_y \quad (1.5)$$

$$\ddot{z} = -n^2z + d_z + a_z \quad (1.6)$$

where $[d_x, d_y, d_z]^T$ and $[a_x, a_y, a_z]^T$ are, respectively, the vectors of environmental perturbations and control accelerations.

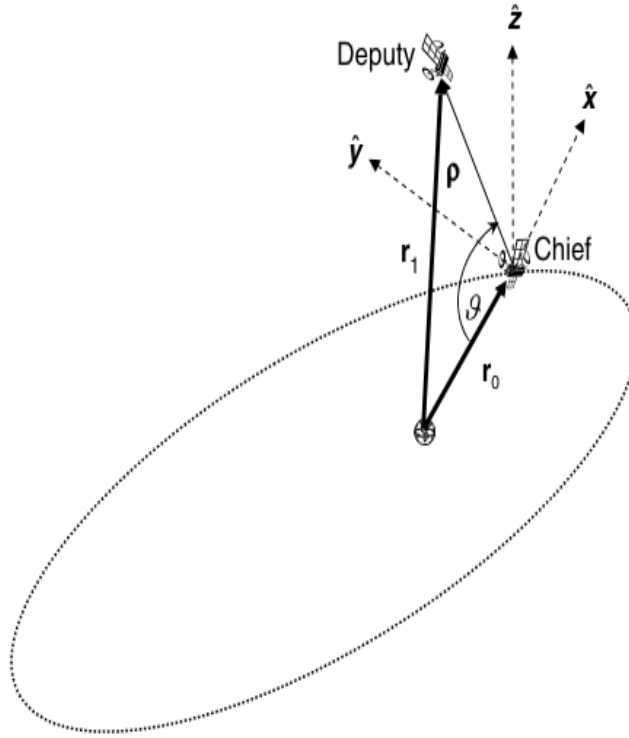


Figure 1: Rotating Euler–Hill frame, centered at the chief spacecraft. This figure also shows the deputy spacecraft, whose position vectors in the rotating and inertial reference frames are denoted by ρ and r_1 , respectively [\[4\]](#)

For ease of analysis, the relative coordinates can be normalized by the radius of the reference orbit, and the angular velocities by n . The normalized, unforced Clohessy-Wiltshire equations (1.1) – (1.3) become

$$\bar{x}'' - 2\bar{y}' - 3\bar{x} = 0 \quad (1.7)$$

$$\bar{y}'' + 2\bar{x}' = 0 \quad (1.8)$$

$$\bar{z}'' + \bar{z} = 0 \quad (1.9)$$

where $(\cdot)'$ denotes differentiation with respect to normalized time. linearizing differential equations (1.1) – (1.3) in state-space form. Defining the state vector $\mathbf{x} = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^T$, equations (1.1) – (1.3) assume the form

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) \quad (1.10)$$

Where $\mathbf{A}$ is the system matrix, given by

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix} \quad (1.11)$$

and the initial conditions are $\mathbf{x}(t_0) = [x(t_0), y(t_0), z(t_0), \dot{x}(t_0), \dot{y}(t_0), \dot{z}(t_0)]^T$. The eigenvalues of $\mathbf{A}$ are $\{\pm nj, \pm nj, 0, 0\}$, so a secular mode is expected to appear in the solution. The solution to these equations may be formulated in terms of the state transition matrix, $e^{\mathbf{A}(t-t_0)}$,

$$\mathbf{x}(t) = e^{\mathbf{A}(t-t_0)} \mathbf{x}(t_0) \quad (1.12)$$

Following the usual steps for computing the transition matrix yields (for $t_0 = 0$),

$$e^{At} = \begin{bmatrix} 4 - 3c_{nt} & 0 & 0 & \frac{s_{nt}}{n} & \frac{2}{n} - \frac{2s_{nt}}{n} & 0 \\ -6nt + 6s_{nt} & 1 & 0 & -\frac{2}{n} + \frac{2c_{nt}}{n} & \frac{4s_{nt}}{n} - 3t & 0 \\ 0 & 0 & c_{nt} & 0 & 0 & \frac{s_{nt}}{n} \\ 3ns_{nt} & 0 & 0 & c_{nt} & 2s_{nt} & 0 \\ -6n + 6nc_{nt} & 0 & 0 & -2s_{nt} & -3 + 4c_{nt} & 0 \\ 0 & 0 & -ns_{nt} & 0 & 0 & c_{nt} \end{bmatrix} \quad (1.13)$$

We can now determine, by substituting Eq. (1.13) into Eq. (1.12), the solutions to the relative position and velocity components:

$$x(t) = \left[4x(0) + \frac{2\dot{y}(0)}{n} \right] + \frac{\dot{x}(0)}{n} \sin(nt) - \left[3x(0) + \frac{2\dot{y}(0)}{n} \right] \cos(nt) \quad (1.14)$$

$$y(t) = -[6nx(0) + 3\dot{y}(0)]t + \left[y(0) - \frac{2\dot{x}(0)}{n} \right] + \left[6x(0) + \frac{4\dot{y}(0)}{n} \right] \sin(nt) + \frac{2\dot{x}(0)}{n} \cos(nt) \quad (1.15)$$

$$z(t) = \frac{\dot{z}(0)}{n} \sin(nt) + z(0) \cos(nt) \quad (1.16)$$

$$\dot{x}(t) = \dot{x}(0) \cos(nt) + [3x(0)n + 2\dot{y}(0)] \sin(nt) \quad (1.17)$$

$$\dot{y}(t) = -[6nx(0) + 3\dot{y}(0)] + [6x(0)n + 4\dot{y}(0)] \cos(nt) - 2\dot{x}(0) \sin(nt) \quad (1.18)$$

$$\dot{z}(t) = \dot{z}(0) \cos(nt) - z(0)n \sin(nt) \quad (1.19)$$

Firstly, the linear system (1.10) has multiple equilibria $\{x = 0, y = \text{constant}, z = 0\}$. Secondly, it's observed that through linearization, the movement perpendicular to the plane (out-of-plane) has been separated from the movement within the plane (in-plane). This perpendicular movement follows a simple harmonic pattern, which means its effect can be eliminated by carefully choosing the starting conditions: $z(t) = \dot{z}(0) = 0$. Lastly, the transverse, along-track, component indicates the existence of **drift**, varying linearly with time, implying the in-plane motion is unstable. This is shown due to the term $y(t)$ going to infinity as t goes to infinity. Nevertheless, a stable subspace within the state-space can be identified, thereby rendering the motion stable. The initial conditions which span the stable subspace are given by

$$\dot{y}(0) = -2nx(0) \quad (1.20)$$

Choosing initial conditions (1.20) will yield a bounded relative motion to first order, i.e., only for the Clohessy-Wiltshire model. It is important to note that the concept of stability in the context of the Clohessy-Wiltshire equations is local.

2.2. Control algorithm: Formulation of equations

To maintain the formation with artificial potentials, consider the attractive potential,

$$U = \frac{1}{2}k(\vec{\mathbf{r}} - \vec{\mathbf{r}}_d)^2 \quad (1.21)$$

Where K is the strength of potential field, $\vec{\mathbf{r}}$, the position of our satellite and $\vec{\mathbf{r}}_d$, the desired equilibrium position

The force induced by this potential

$$\vec{\mathbf{F}}_{goal} = -\nabla U \quad (1.22)$$

adding these virtual control forces to the right part of Clohessy-Wiltshire equation

$$\ddot{x} = 3n^2x + 2n\dot{y} + \frac{\partial U}{\partial x} \quad (1.23)$$

$$\ddot{y} = -2n\dot{x} + \frac{\partial U}{\partial y} \quad (1.24)$$

$$\ddot{z} = -n^2z + \frac{\partial U}{\partial z} \quad (1.25)$$

where

$$\frac{\partial U}{\partial x} = -kx \left(1 - \frac{r_d}{r}\right) \quad (1.26)$$

$$\frac{\partial U}{\partial y} = -ky \left(1 - \frac{r_d}{r}\right) \quad (1.27)$$

$$\frac{\partial U}{\partial z} = -kz \left(1 - \frac{r_d}{r}\right) \quad (1.28)$$

Substituting in the equation (1.23) – (1.24), we get the following

$$\ddot{x} = 3n^2x + 2n\dot{y} - kx \left(1 - \frac{r_d}{r}\right) \quad (1.29)$$

$$\ddot{y} = -2n\dot{x} - ky \left(1 - \frac{r_d}{r}\right) \quad (1.30)$$

$$\ddot{z} = -n^2 z - kz \left(1 - \frac{r_d}{r}\right) \quad (1.31)$$

In order to achieve a stable equilibrium point, some virtual friction will be added.

Virtual friction compensation of free terms in Clohessy-Wiltshire equations

$$\ddot{x} = -f_r e_r \frac{x}{|r|} - f_r (v_{tx} - v_{txd}) \quad (1.32)$$

$$\ddot{y} = -f_r e_r \frac{y}{|r|} - f_r (v_{ty} - v_{tyd}) \quad (1.33)$$

$$\ddot{z} = -f_r e_r \frac{z}{|r|} - f_r (v_{tz} - v_{tzd}) \quad (1.34)$$

where,

$$e_r = \mathbf{r} \cdot \dot{\mathbf{r}} \quad (1.35)$$

$$\mathbf{v}_\tau = \dot{\mathbf{r}} - e_r \frac{\mathbf{r}}{|r|} \quad (1.36)$$

$$\mathbf{v}_{\tau d} = \alpha \frac{\mathbf{v}_\tau}{|v_\tau|} \quad (1.37)$$

where α = control design coefficient, f_r = friction coefficient

To prevent collisions, a repulsive potential is induced by each obstacle denoted as (B). Each obstacle has a defined radius of influence, denoted as (d), which extends from the center of the obstacle outward within a spherical boundary of radius (d). The obstacle generates a repulsive potential field that affects all satellites and increases the closer they are to the obstacle. The potential calculated using,

$$u_R = \frac{k_r}{2d(q,B)^2} \quad (1.38)$$

Where $k_r > 0$ is a scaling factor. The potential is only properly defined for points outside the obstacle $d(q, B) > 0$. The force induced by the obstacle potential is

$$\vec{F}_R = \frac{k}{d^3(q,B)} \frac{\partial d}{\partial q} \quad (1.39)$$

Summing all repulsive forces in a deputy's vicinity, this yields total repulsive force,

$$\vec{F}_{total R} = \sum_i^n \vec{F}_R \quad (1.40)$$

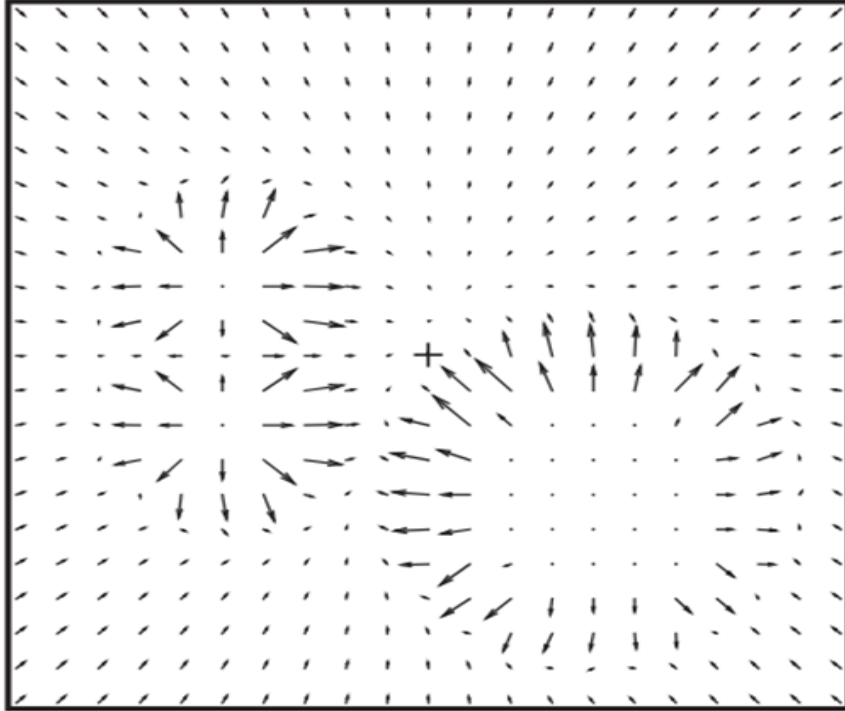


Figure 2: Forces induced by the potential function [1]

Add the potentials and fiction to the right-hand side of the Clohessy-Wiltshire equations, we get the control, the equation are as follows,

$$\ddot{x} = 3n^2x + 2n\dot{y} - kx \left(1 - \frac{r_d}{r}\right) - f_r e_r \frac{x}{|r|} - f_r (v_{\tau x} - v_{\tau xd}) + \sum_i^n F_{Rx} \quad (1.41)$$

$$\ddot{y} = -2n\dot{x} - ky \left(1 - \frac{r_d}{r}\right) - f_r e_r \frac{y}{|r|} - f_r (v_{\tau y} - v_{\tau yd}) + \sum_i^n F_{Ry} \quad (1.42)$$

$$\ddot{z} = -n^2z - kz \left(1 - \frac{r_d}{r}\right) - f_r e_r \frac{z}{|r|} - f_r (v_{\tau z} - v_{\tau zd}) + \sum_i^n F_{Rz} \quad (1.43)$$

In addition to the above control equations (1.41) – (1.43). consider another variation of the harmonic potential,

$$U(r) = \frac{1}{2} (\vec{r} - \vec{r}_{goal})^T k (\vec{r} - \vec{r}_{goal}) \quad (1.46)$$

Where, $\vec{r}_{goal}$ is the target position, k is strength. The force indices by this potential are

$$\vec{F}_{goal}(r) = -\frac{\partial U_{goal}}{\partial r} = k(\vec{r} - \vec{r}_{goal}) \quad (1.47)$$

An attractive force proportional to the distance from the goal. With the potential from equation (1.21), a desired distance, from that target can be specified and the deputy tries to keep this desired distance, r_d . With the potential from (1.46), the deputy can move directly to the goal position avoiding obstacles. The Control equations (1.41) – (1.43) are updated by changing the attractive potential (1.21) with (1.46) . Hence the new control equations are as follows,

$$\ddot{x} = 3n^2x + 2n\dot{y} - k(r_x - r_{goal.x}) - f_r e_r \frac{x}{|r|} - f_r(v_{tx} - v_{txd}) + \sum_i^n F_{Rx}$$

(1.48)

$$\ddot{y} = -2n\dot{x} - k(r_y - r_{goal.y}) - f_r e_r \frac{y}{|r|} - f_r(v_{ty} - v_{tyd}) + \sum_i^n F_{Ry}$$

(1.49)

$$\ddot{z} = -n^2z - k(r_z - r_{goal.z}) - f_r e_r \frac{z}{|r|} - f_r(v_{tz} - v_{tzd}) + \sum_i^n F_{Rz}$$

(1.50)

For numerical analysis, when studying the divergence and convergence of this system under control, A parameter called Dispersion, D is used. This is the sum of distances which gives a sense of how spread-out the models are in space. For a non convergent and non divergent system, this Dispersion value is to be both bounded above and below. This value is calculated using the formular,

$$\sum_{i \neq j}^n \|\vec{r}_i - \vec{r}_j\| = D \quad (1.44)$$

Where, $\vec{r}$ is position of a Satellite and D is the dispersion amount.

In addition to this, Monitoring the fuel consumed by the model swarm implies the efficiency and also viability of this system. Fuel consumption in many systems (especially in vehicles, drones, or any moving machinery) is closely related to the amount of acceleration. Higher acceleration typically requires more energy, hence more fuel. Although s_k (fuel consumed) is not a direct measure of the mass of fuel

consumed, it is proportional to the energy required to produce the acceleration over the given time interval, which in turn is proportional to the mass of fuel consumed. This proportionality ensures that the fuel consumption s_k can be approximately monitored and related to the efficiency of the system.

$$s_k = \frac{\Delta t}{2} (||\mathbf{u}_k|| + ||\mathbf{u}_{k+1}||) \quad (1.45)$$

Where ,

- Δt This represents the difference in time between two consecutive time steps k and $k+1$
- $||\mathbf{u}_k||$ and $||\mathbf{u}_{k+1}||$ is the magnitude of Acceleration at time step k and magnitude of Acceleration at time step $k+1$ respectively,
- s_k ,the measure is the fuel consumed (ΔV) by the object during the time interval from t_k to t_{k+1}

2.3. Control algorithm: Structure

For the first approach to the control algorithm implementation, the deputy satellites in the model system are divided into groups based on two parameters: the distance between them and the drift value from equation (1.20). These groups consist of deputy model satellites with a distance less than a specific value ε_1 and a drift less than another parameter ε_2 . An **adjacency matrix** is used to store this relationship between satellites. If both conditions are satisfied, we add a 1 to this matrix else a 0. This resulting in three types of groups: the **largest group** (with the highest number of deputies), the **medium group**, and **the single satellite deputy group**. To minimize fuel consumption, both the **single satellite deputy group** and the **medium group** are attracted to their closest point in the **largest group** using an attractive potential respectively. In addition, each object exerts a repulsive force on all other deputies in its vicinity. Finally, in groups with more than one satellite, an attractive potential is used to maintain the group and distance from each other. The adjacency matrix for a system of 20 model satellites is show in Figure [3] below.

	s1	s2	s3	s4	s5	s6	s7	s8	s9	s10	s11	s12	s13	s14	s15	s16	s17	s18	s19	s20
s1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s3	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	1	1	0	0
s4	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0
s5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s8	0	0	1	0	0	0	0	0	0	0	0	0	0	1	1	0	1	1	0	0
s9	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
s10	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
s11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s12	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s14	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	1	0	0
s15	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0
s16	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s17	0	0	1	0	0	0	0	1	0	0	0	0	0	1	0	0	0	1	0	0
s18	0	0	1	0	0	0	0	1	0	0	0	0	0	1	1	0	1	0	0	0
s19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 3: Adjacency matrix of a System of 20 model satellites

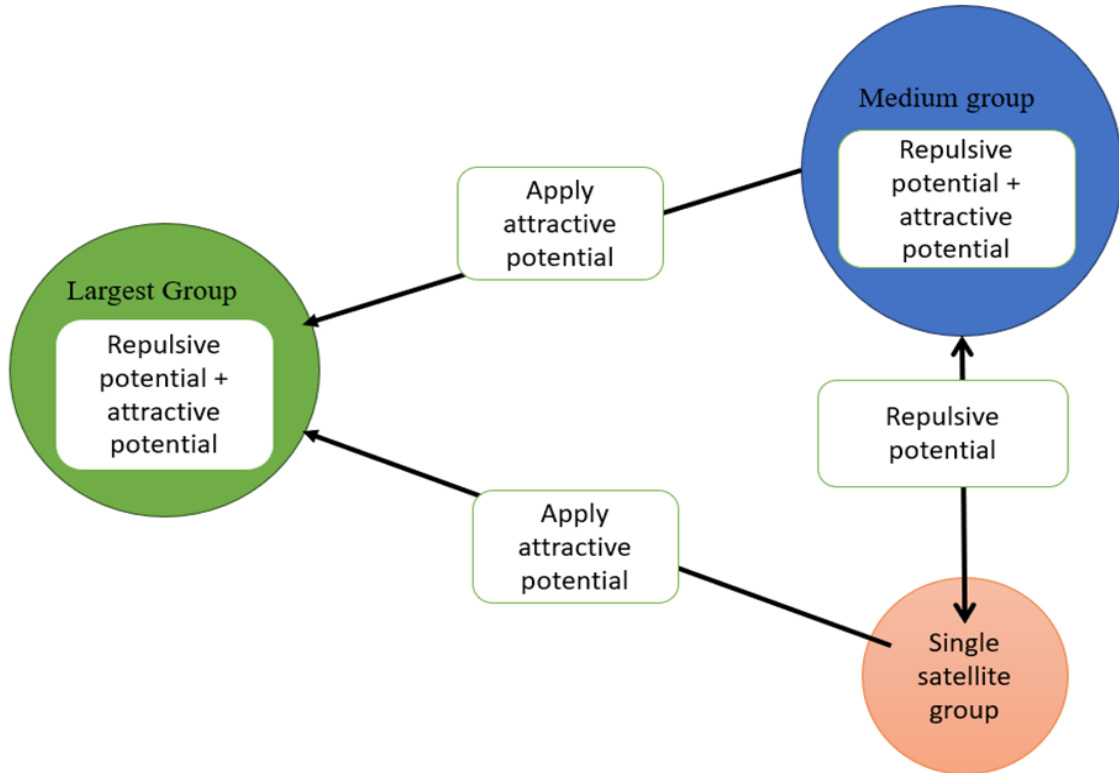


Figure 4: Schematics of the First Approach: Control Algorithm Using Artificial Potentials

Figure [4] demonstrates the application of artificial potentials in controlling the satellite swarm. The largest group employs a combination of both attractive and repulsive potentials to maintain its shape and avoid collisions with each satellite within the group. Similarly, the medium group uses these potentials to preserve its formation, with an additional attractive potential drawing it towards the largest group. Lastly, the single satellite group is influenced by an attractive potential that also directs it towards the largest group. In the second approach to the control algorithm implementation, all model satellites have an attractive potential on them, with the system of satellites orbit as the attractive potential center, simultaneously, each satellite has a repulsive potential, and it self as its repulsive potential center. This approach makes use of the control equations [\(1.41\)](#) – [\(1.43\)](#)

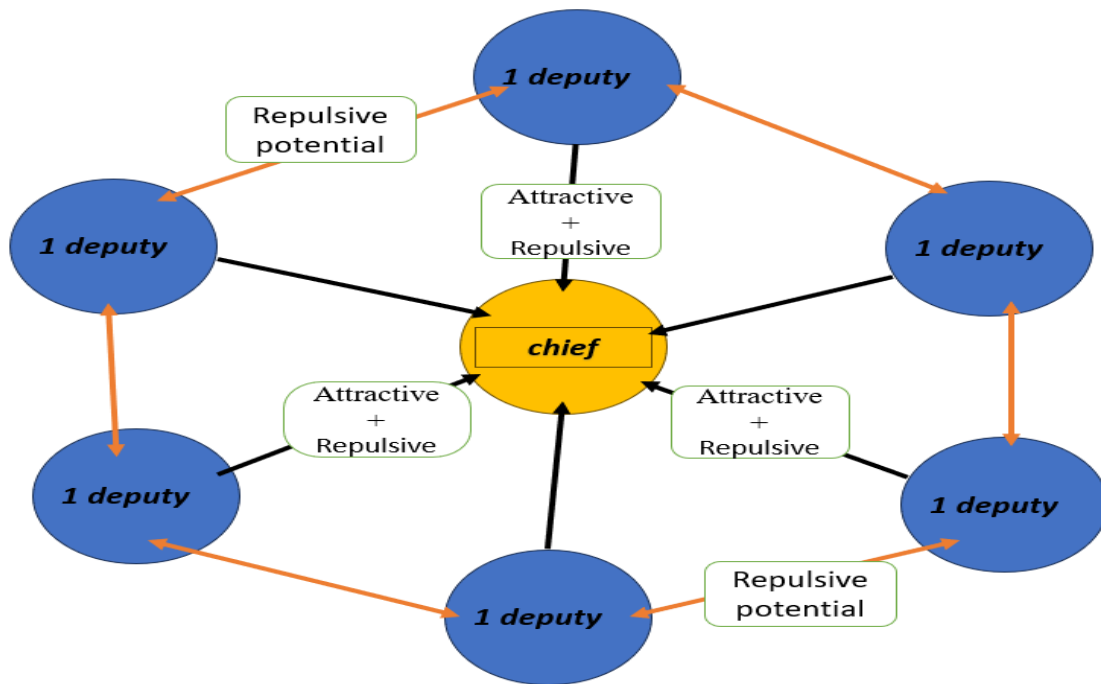


Figure 5: Schematics of the Second Control Approach

Figure [5] depicts the application of these potentials in our control system. All deputy satellites are influenced by an attractive potential that moves them to a desired distance of 10 meters from the systems of satellites orbit. Simultaneously, each deputy satellite experiences a repulsive potential to avoid collisions with all other deputy satellites.

3. Results of simulation

For the simulation, a system of 20 model satellites with random positions and velocities is used. Each model satellite is assigned a random position and velocity, which are stored in a file. These same random positions and velocities are utilized for both control approaches to ensure consistency and comparability in evaluating their performance.

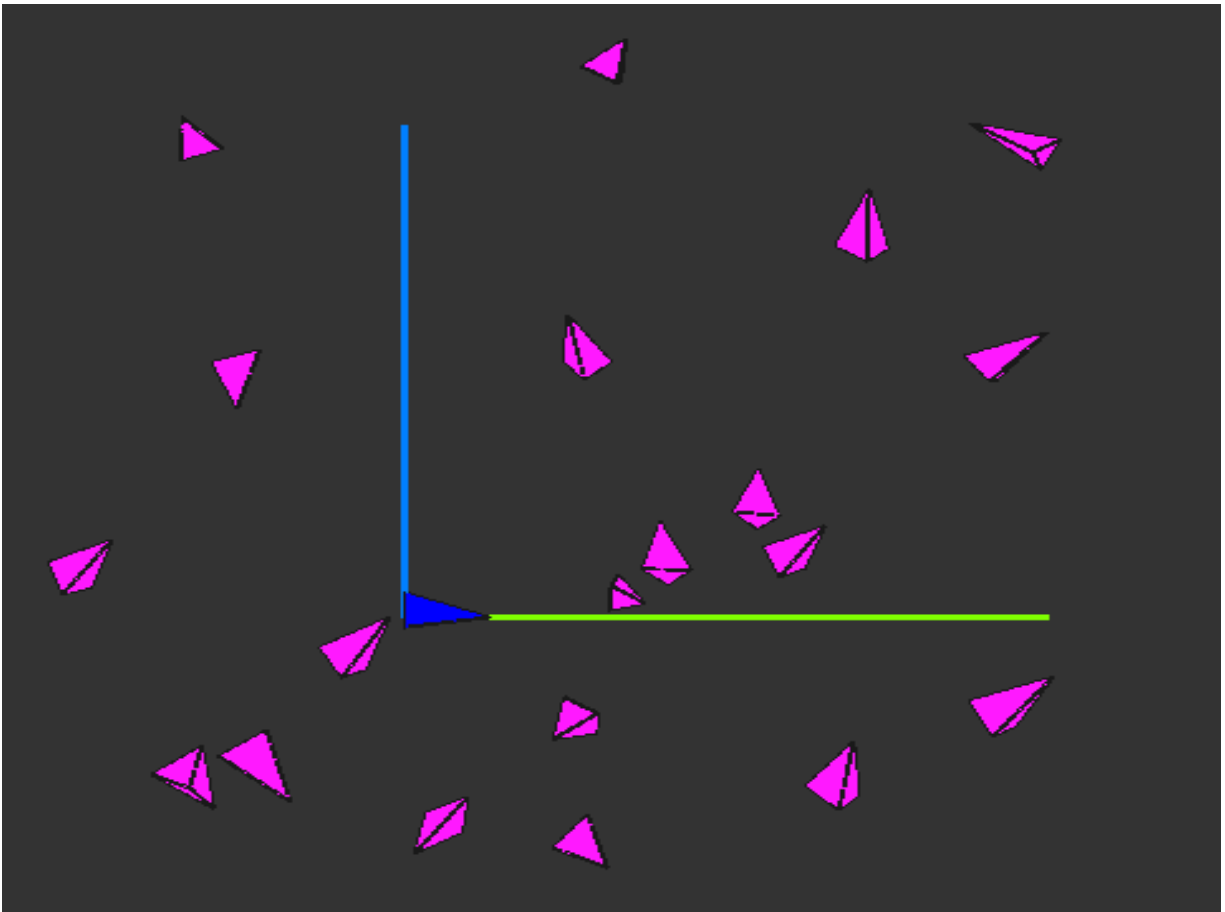


Figure 6: model system of 20 satellites with random velocities and positions indicated by the pink pyramids

Figure [6] depicts a model swarm system of 20 satellites indicated by the pink pyramids, with the blue pyramid representing the orbit which the system of model satellites is to follow. Each satellite is assigned a random position and velocity. These initial conditions are used throughout the study to examine in detail how each control method and the control coefficients affect the system.

3.1. Simulation with no control Results

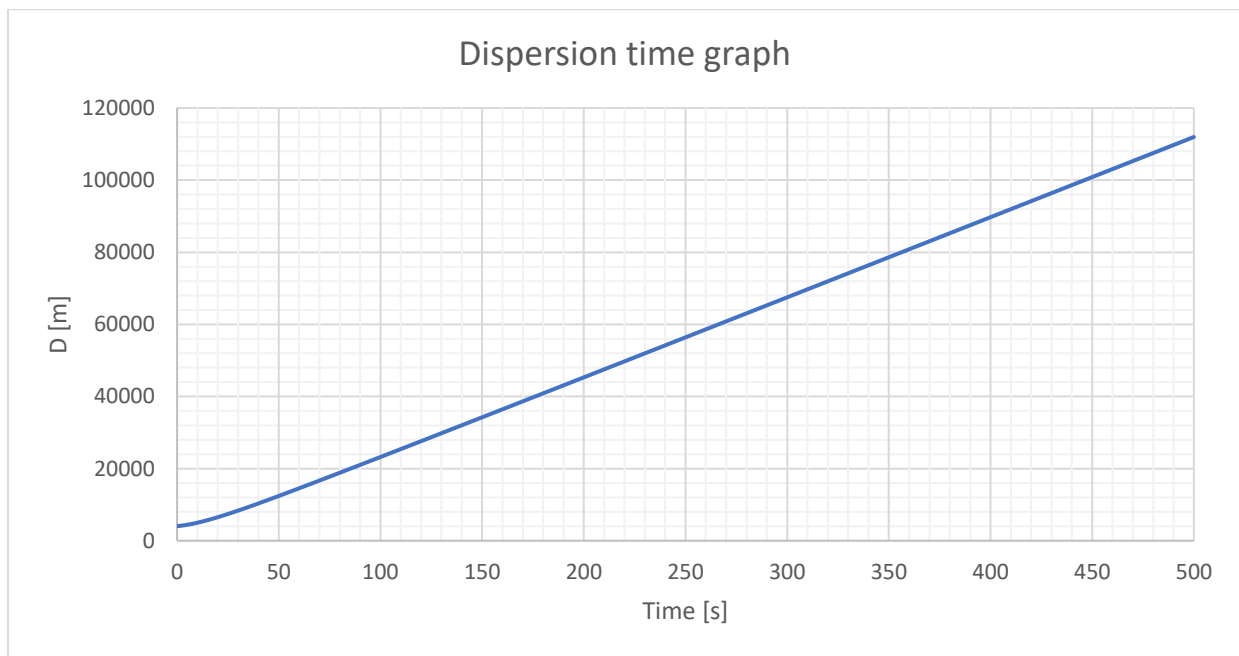


Figure 7: Dispersion Time graph of a system of 20 satellites with no control.

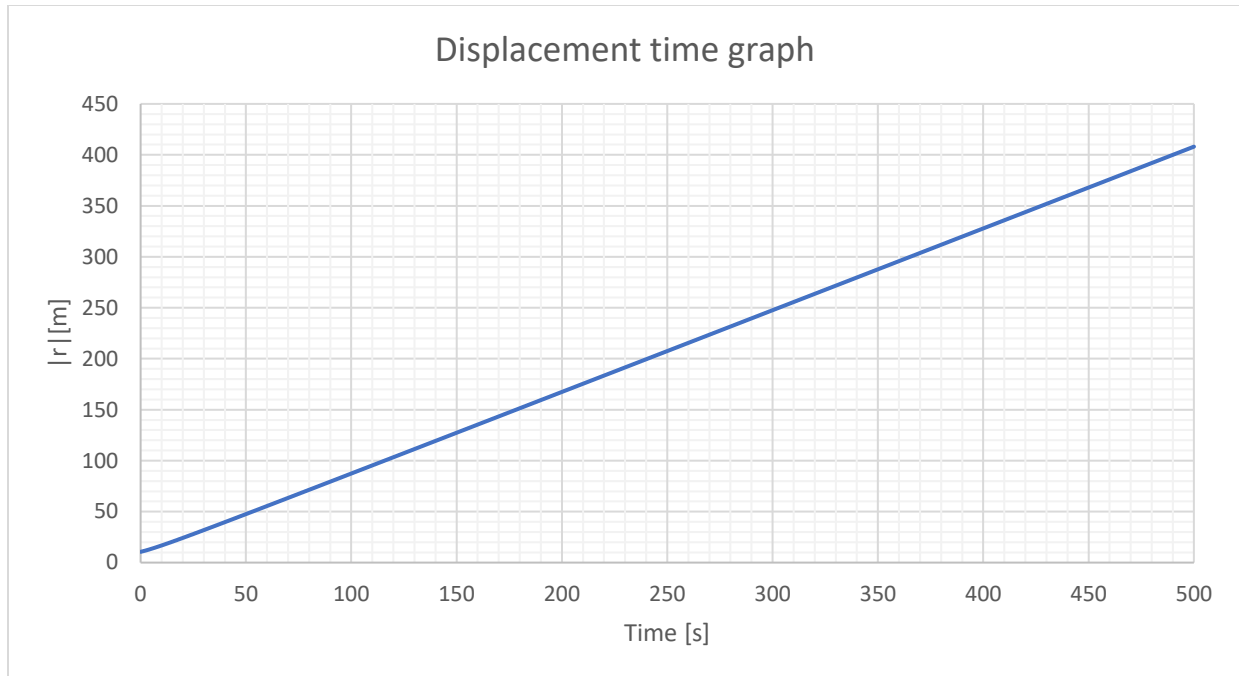


Figure 8: Displacement Time Graph of one satellite from the system of 20 with no control.

From Figure [7], The dispersion, D , is observed to have no upper bound, it diverges. The same behaviour is seen in Figure [8], where the displacement time graph of on the satellite models in the system, diverges to infinity.

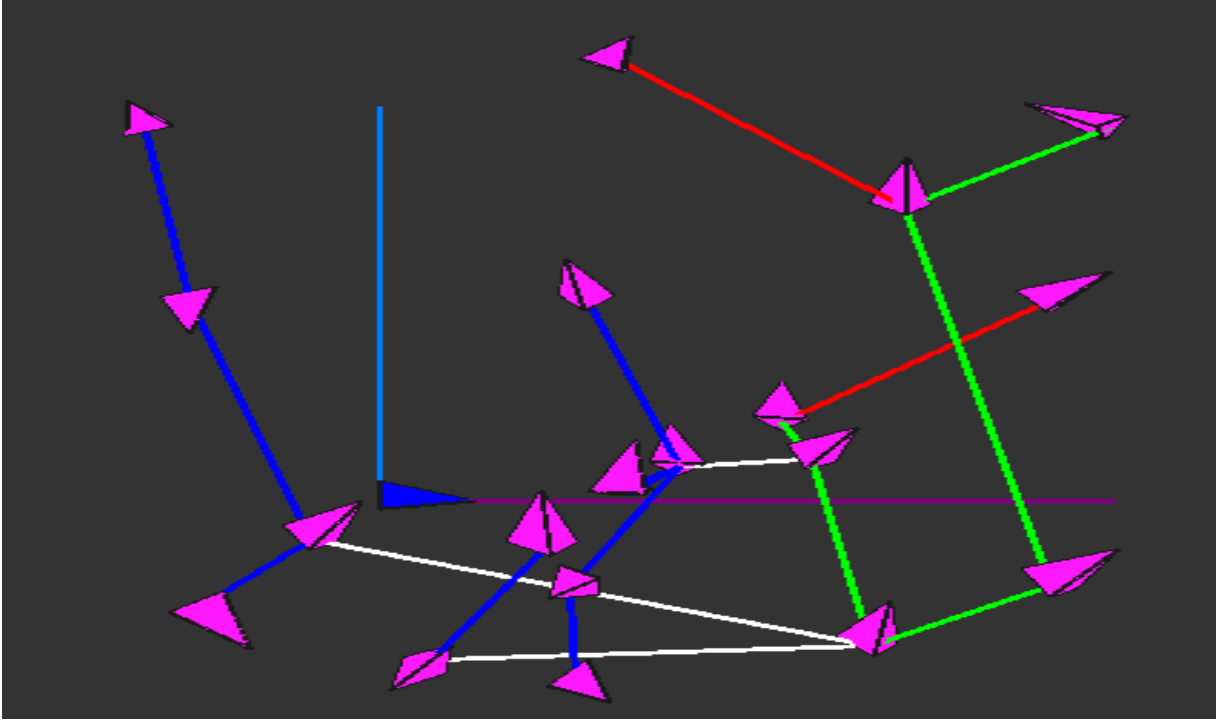


Figure 9: Model of system of 20 satellites with First Control Approach implemented before simulation. It consists of 3 medium groups, 1 large group, and 2 single satellite groups.

In Figure [9], the blue pyramid represents the known orbit of the system of model satellites, while the pink satellites are identified as the deputies. Three types of satellite groups are represented using different colored lines. The largest group, consisting of the highest number of satellites, is connected by green lines. Satellites linked by the blue line have numbers greater than 0 but fewer than those in the largest group. Individual satellites not part of any group are connected to the nearest satellite in the largest group by the red line, while the nearest satellite in the blue group is connected to the nearest one in the largest group by the white line. According to the

first approach algorithm, the largest group is attempted to preserve its shape. Similarly, the blue group strives to maintain its form, yet satellites within the blue group are attracted towards those in the largest group, as indicated by the white line. Finally, satellites belonging to the single group are attracted to those in the largest group, as shown by the red line towards those in the largest group, as indicated by the white line. Finally, satellites belonging to the single group are attracted to those in the largest group, as shown by the red line.

3.2. First approach control Implementation simulation results

Now the system of model satellites is subjected to the First control approach implementation with the same random initial conditions (velocity and positions).

The Control Coefficients are as follows:

k [<i>strength of Attractive potential field</i>]	0.3
k_r [<i>scaling factor of repulsive potential</i>]	500
f_r [<i>friction coefficient</i>]	1
α [<i>control design coefficient</i>]	0.3
d [<i>collision radius</i>]	4
minimum drift	6
Desired distance	10
Minimum distance	15

Table 1: Control coefficients and parameters for the first control approach

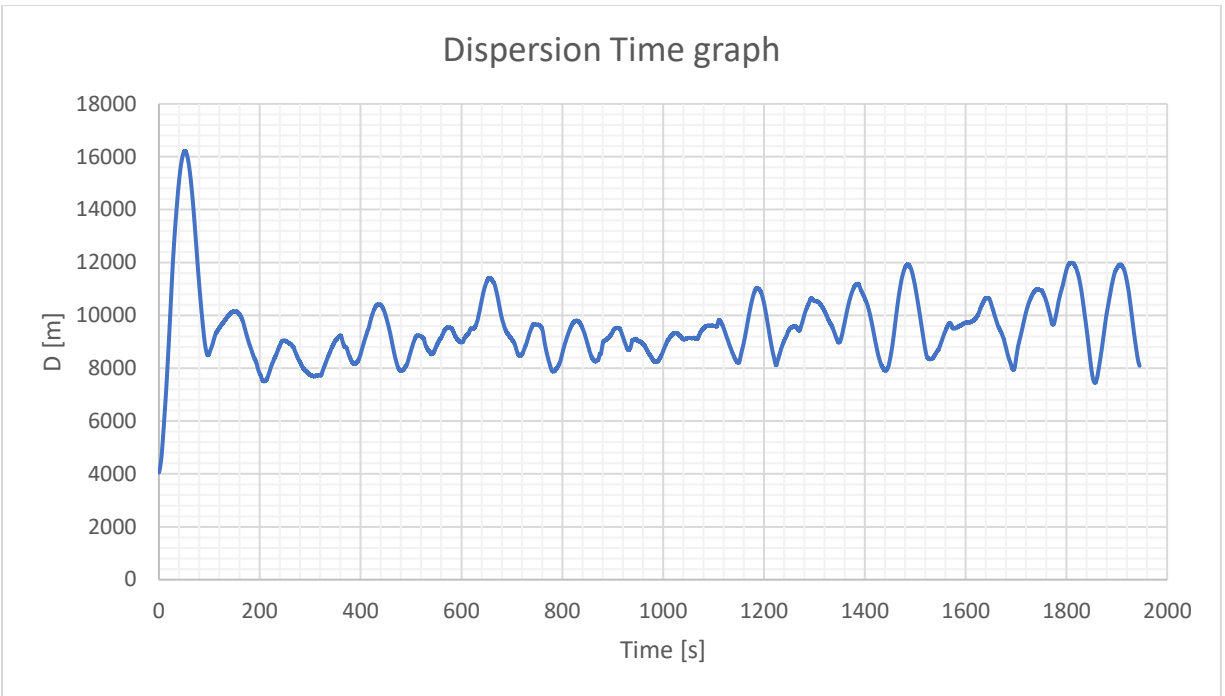
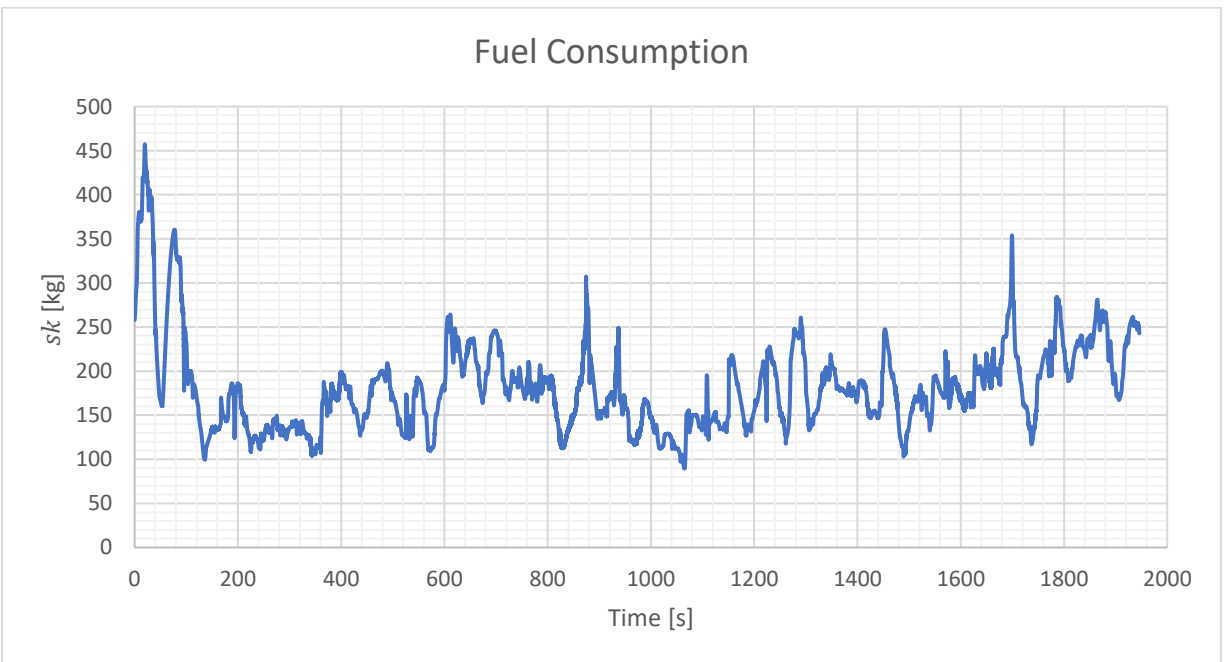
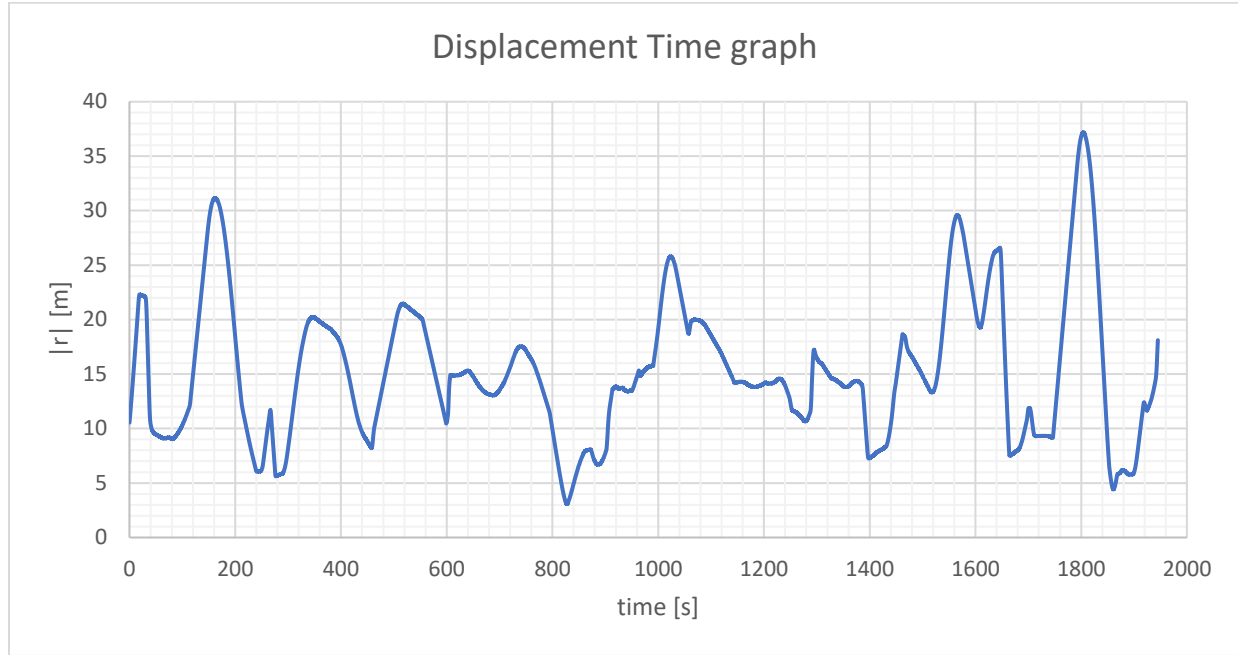


Figure 10: Dispersion time graph of a system of 20 model satellites with bad control



*Figure 11: Fuel consumption time graph of a system of 20 satellites with
bad control*



*Figure 12: Displacement time graph of one satellite from our system of
satellites with bad control*

In Figure [10], the model system does not diverge with the control algorithm implement. Nevertheless, looking at figure [12], there exist the Prescence of unnecessary movement in the motions of the model satellites, which is shown by the oscillating pattern. Updating the coefficients can medicate unnecessary movements.

3.3. First control approach implementation with corrected coefficients results

The movement of the fast-moving models is limited by increasing their friction. The average speed of a model is about 0.5 m/s. A condition is Implement such that if a

velocity exceeds 0.8 m/s, the f_r coefficient for that object is scaled by 2.5. The new test coefficients are as follows:

k	0.2
k_r	500
f_r	1
α	0.3
d	4m
minimum drift	5m/s
minimum distance	10m
desired distance	6m
f_r _scale	2.5
maximum velocity	2 m/s

Table 2: Corrected control coefficients and parameters for the first control approach

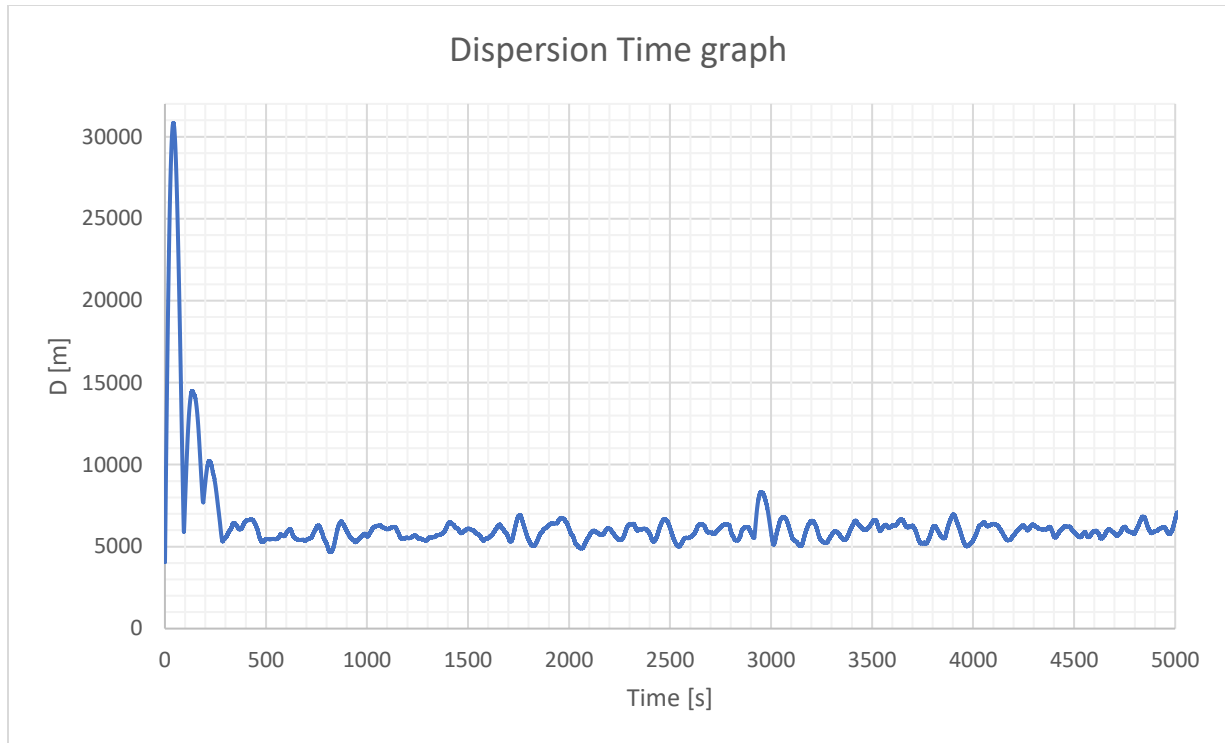


Figure 13: Dispersion time graph for a system of 20 satellites with limiting velocity.

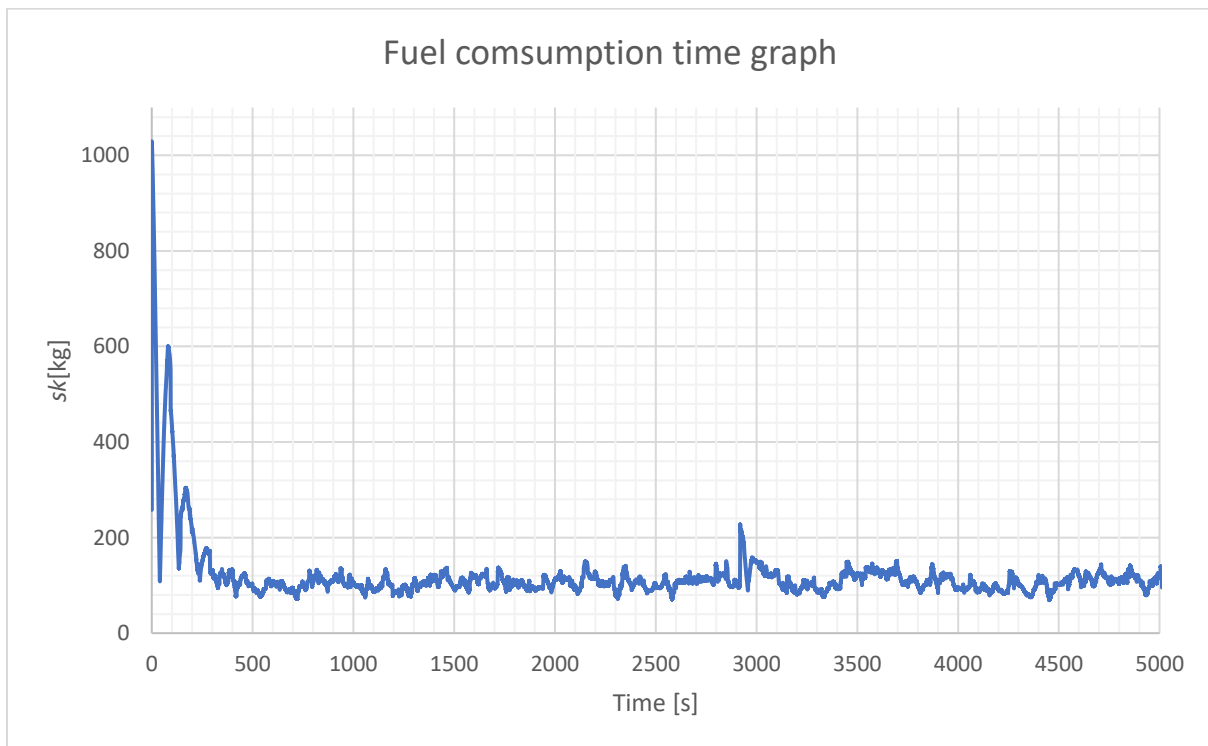


Figure 14: Fuel consumption time graph for a system of 20 satellites with limiting velocity.

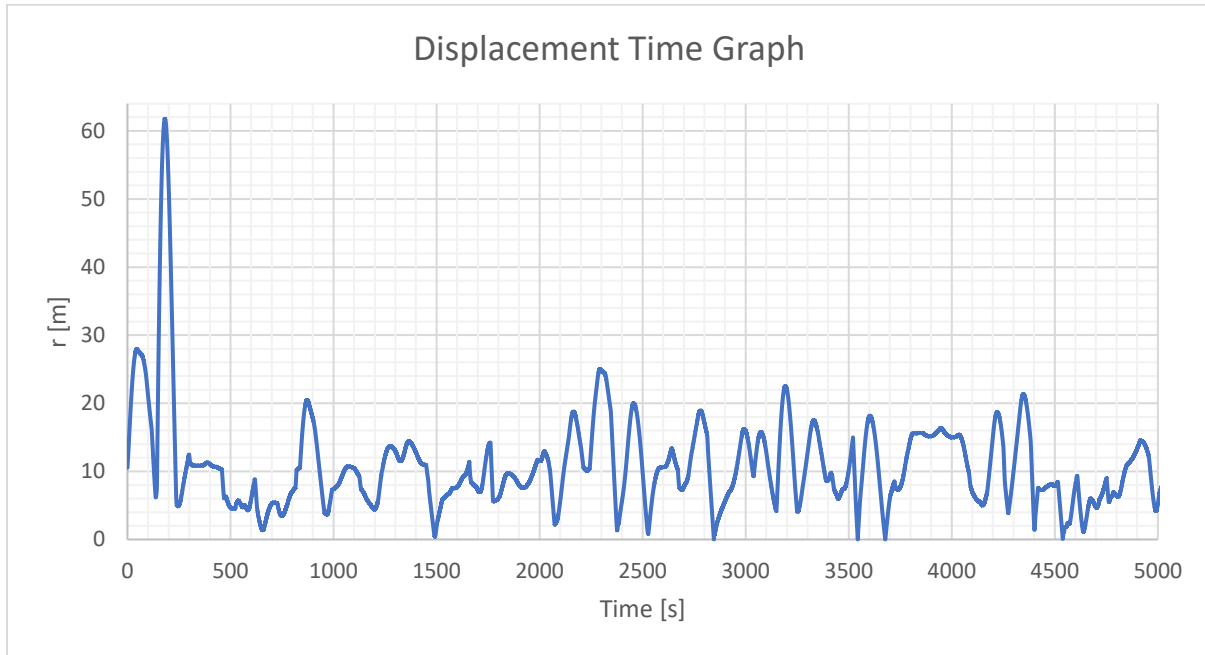


Figure 15: Displacement time one satellite from the system of 20 satellites with limiting velocity.

When comparing both cases, cases with bad and good coefficients,

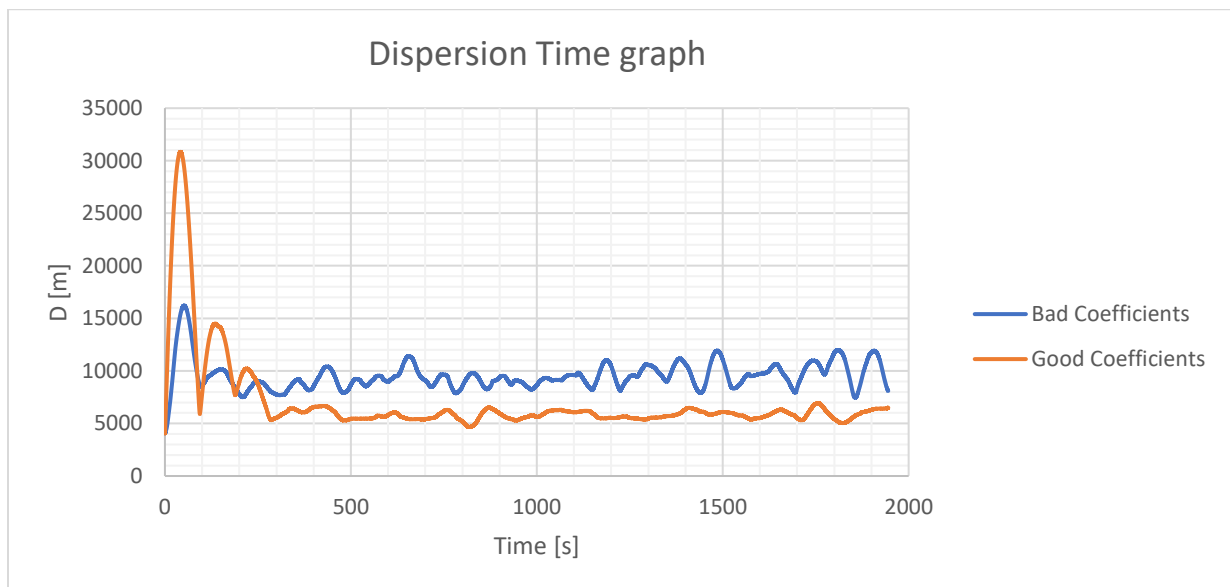


Figure 16: Dispersion time graph comparison with bad coefficients and good coefficients of 20 model satellite system with limiting velocity.

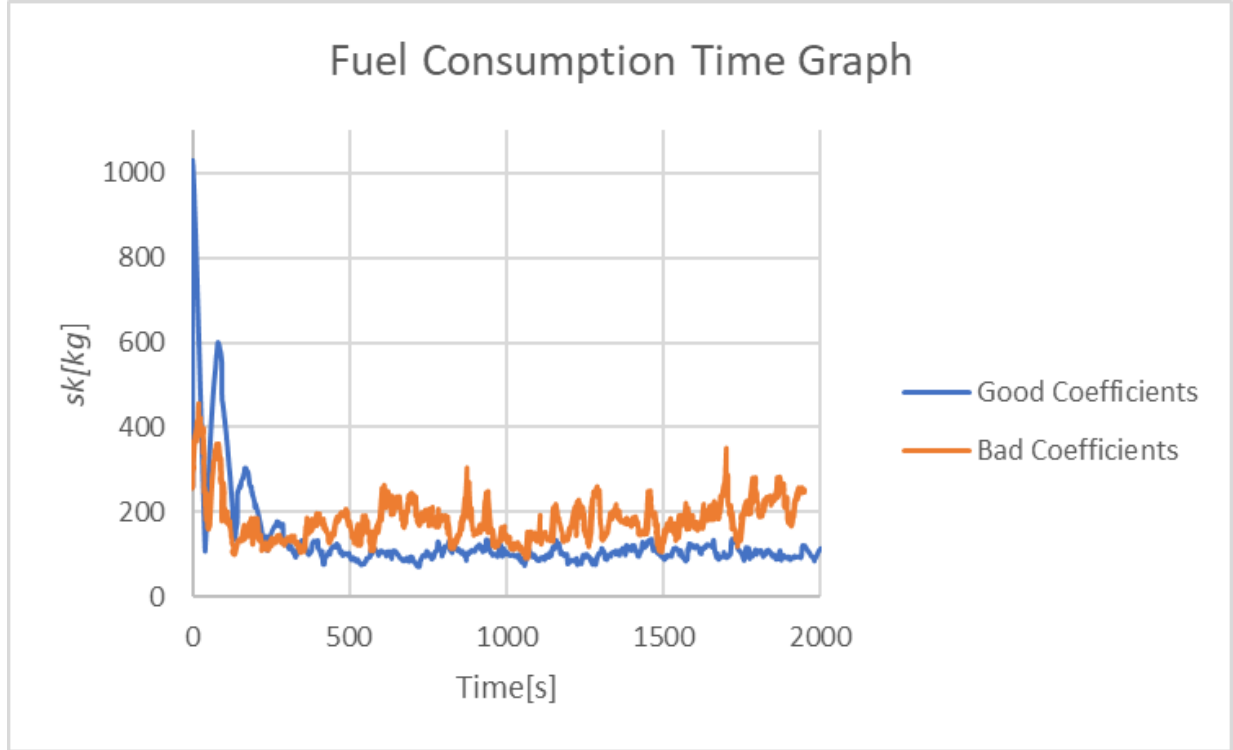


Figure 17: Fuel consumption time graph comparison with bad coefficients and good coefficients of 20 model satellite system with limiting velocity.

Some unnecessary movements have been removed from the model satellite motions, as indicated by figure [15]. In addition, the system is shown to have a lower dispersion value, indicating that the satellites are closer together in a more optimised formation that minimises fuel consumption, while avoiding obstacles. Figure [17] illustrates their fuel consumption comparisons, clearly showing lower fuel consumption with the good coefficients. The control does not specify the shape of the formation but continuously corrects the shape around the vicinity of its orbit

while avoiding obstacles. Figure [18] visually depicts the final formation of the model 20 satellite system.

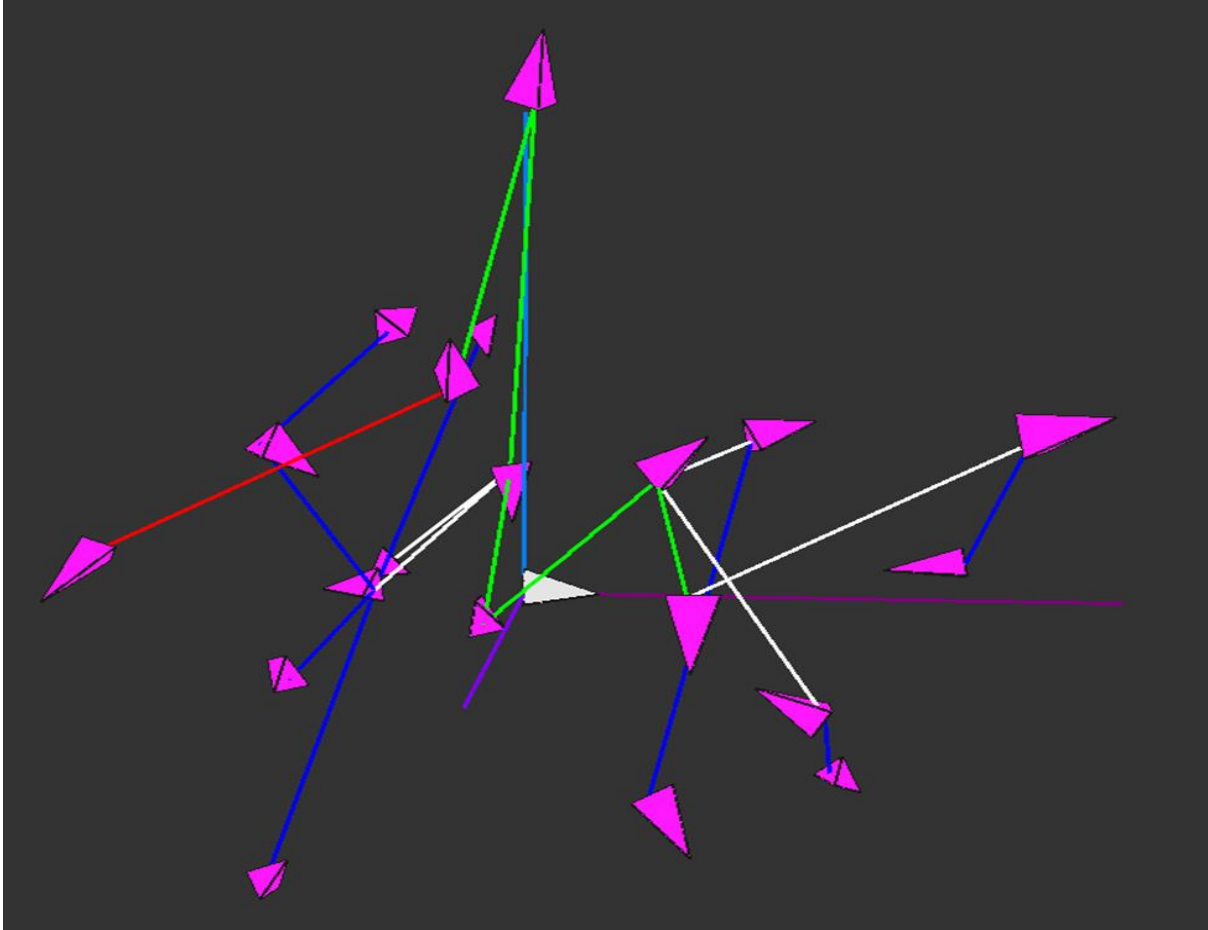


Figure 18: Final shape of a formation of 20 satellites after simulation time 5000 seconds. It consists of 5 medium groups, 1 large group and 1 single satellite group.

3.4. Second approach control implementation simulation results

In section 3.3, the viability of First approach control implementation algorithm has been achieved. This control allows the model satellite system to determine its shape and formation in reaction to its surroundings. Here, the goal is for the formation to take a predefined shape, specifically a sphere, which will later be controlled to lie on a plane.

Considering the sphere has radius 10m, the desired distance coefficient is set to 10m.

The Simulation with new coefficients is set as follows:

k	1
k_r	500
f_r	1
α	0.3
d	4m
desired distance	10m

Table 3: Control coefficients and parameters for the second control approach

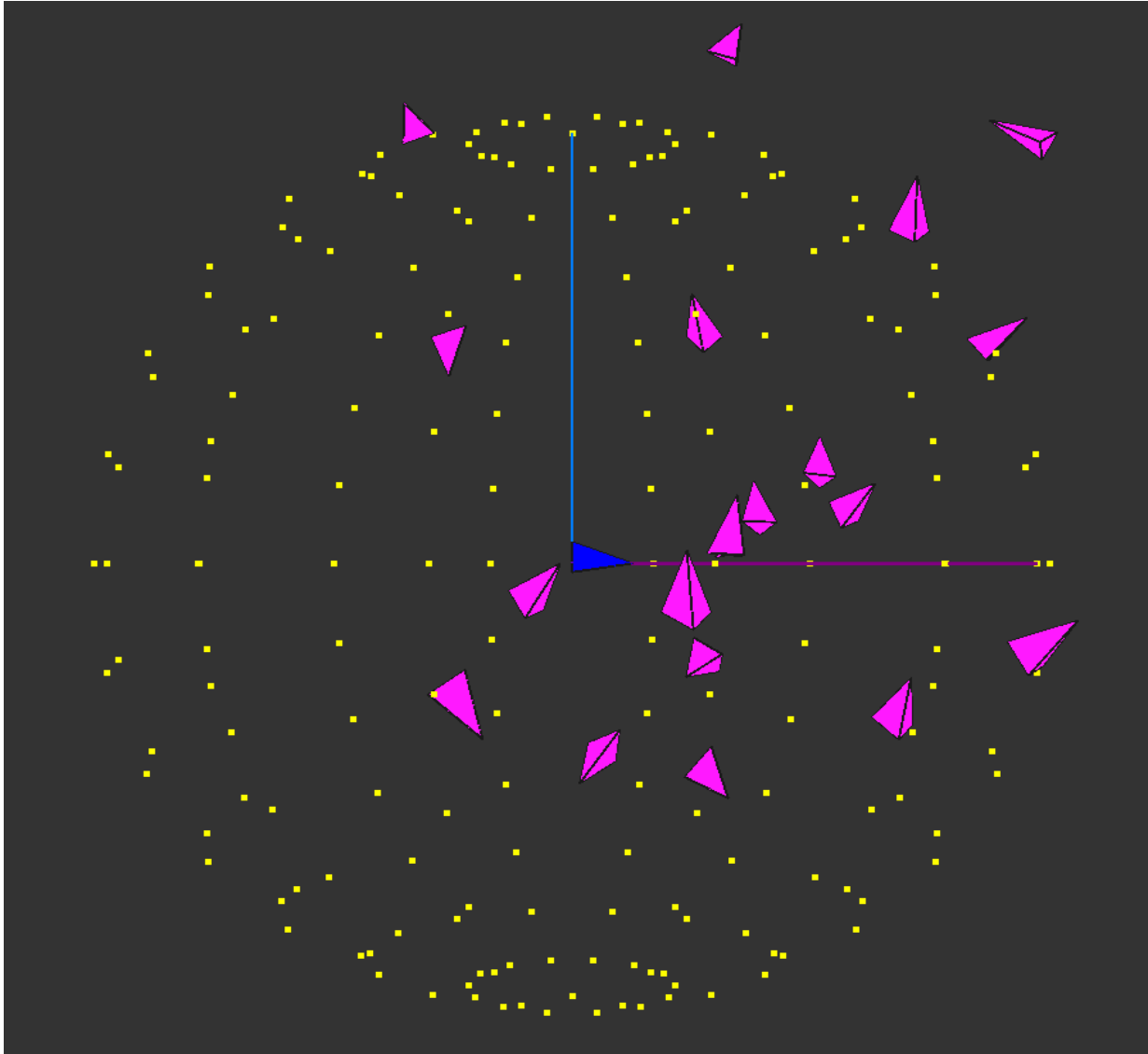


Figure 19: System of 20 model satellites with random positions and velocities before simulation. A sphere of radius 10m is should with small yellow dots indicating the sphere contours. The blue satellite indicates the orbit, and other pink satellites are the deputies

Figure [19] depicts the system of 20 model satellites before simulation. The yellow dots are meant to indicate the surface of a sphere of radius 10m.

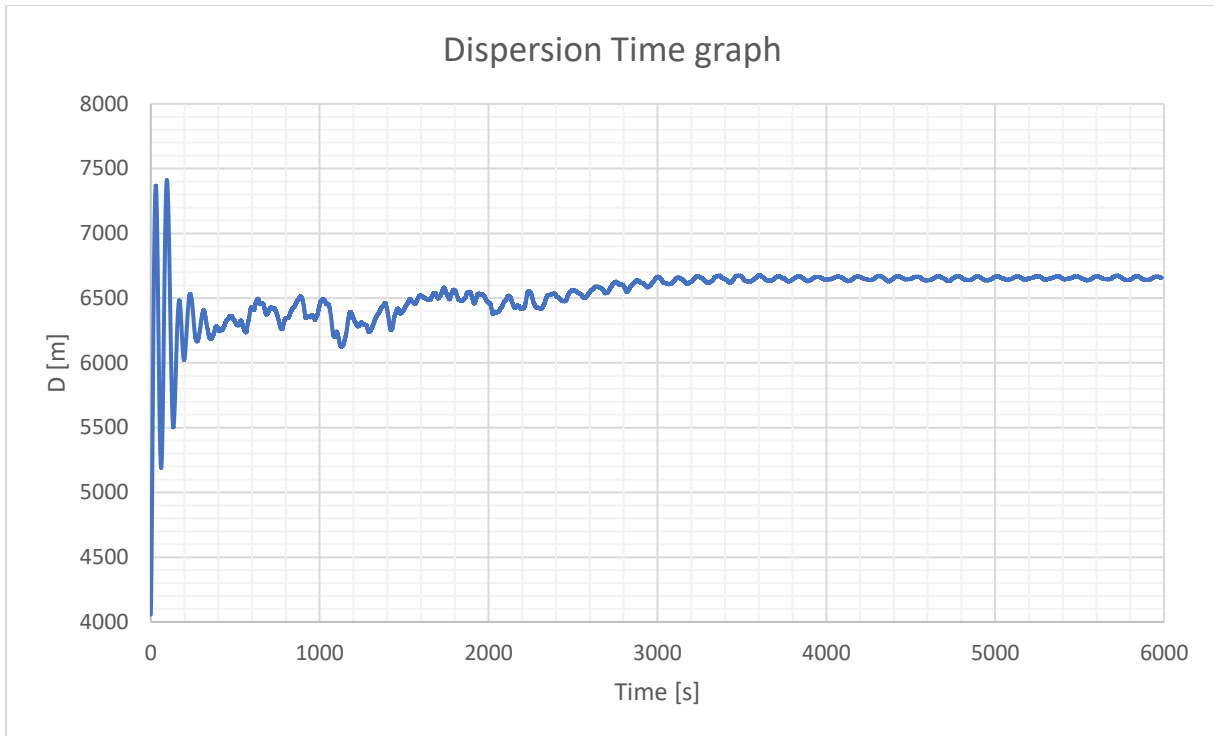


Figure 20: dispersion time graph of a system of 20 satellites

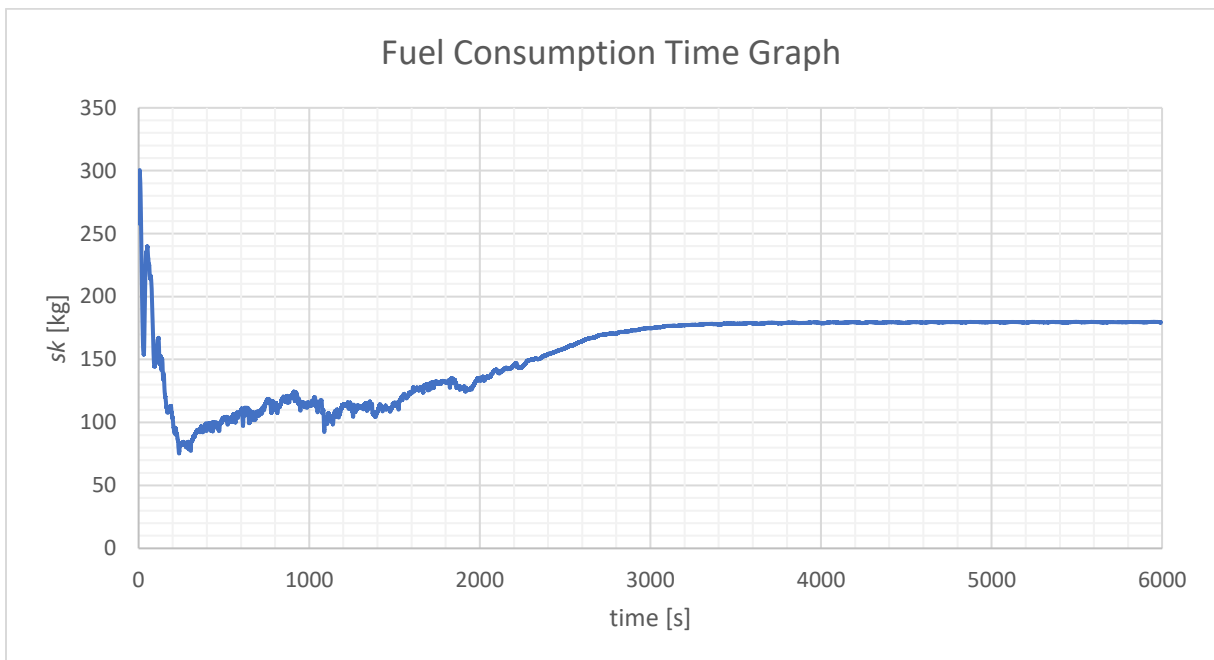


Figure 21: Fuel consumption time graph for a system of 20 satellites

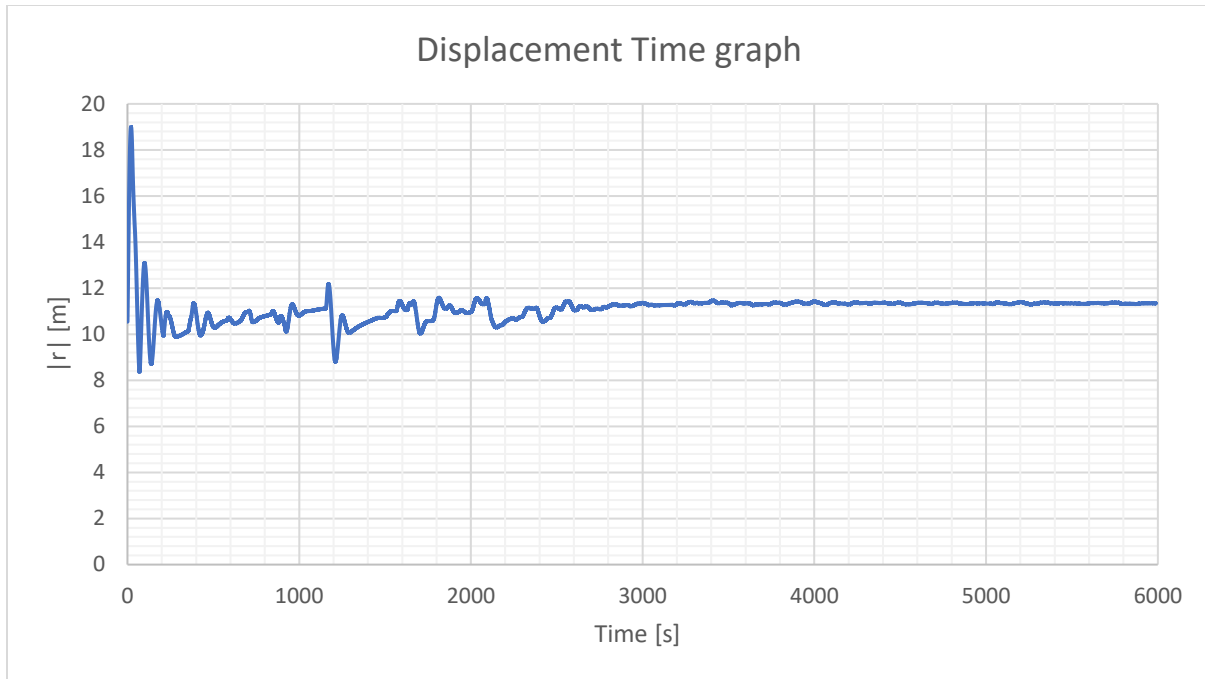


Figure 22: Displacement time one satellite from 20 satellite model

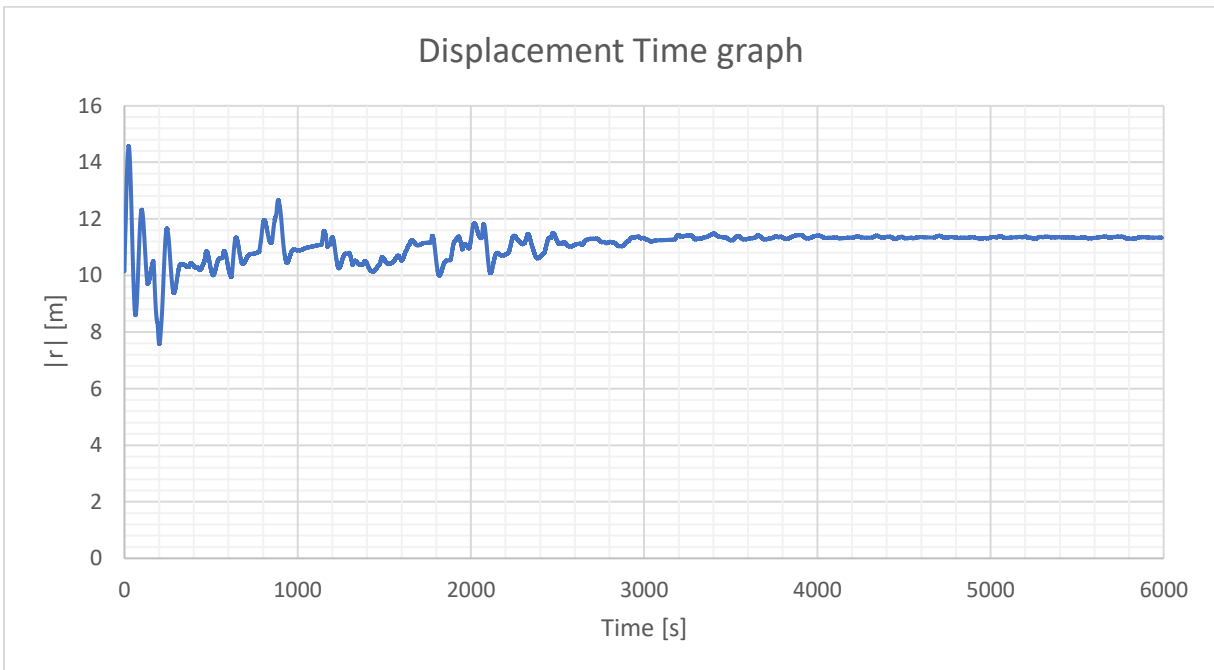


Figure 23: Displacement time one satellite from 20 satellite model

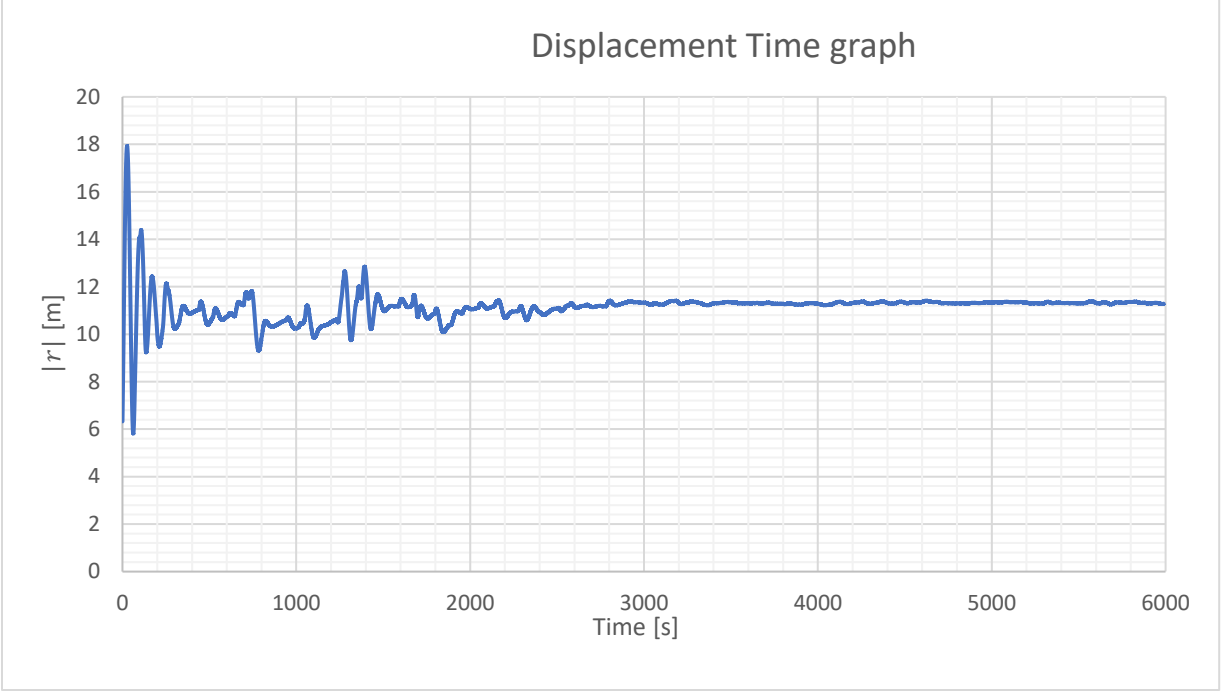


Figure 24: Displacement time one satellite from 20 satellite model

Figure [20], the dispersion time graph indicates the convergence of the dispersion value, proving the viability of the control. Figure [21], after convergence of the dispersion value, the system has constant fuel consumption. Figure [22],[23] and [24] shows a convergence in the displacement Time graphs of 3 different satellites from the model of 20 satellites. The Second Approach control algorithm successfully maintains a sphere-shaped satellite formation. The satellite formation was expected to maintain a distance of 10 m around its orbit, but as shown by the displacement time graphs in Figure [22],[23] and [24], their distances range from 10.8m to 11.5m from the orbit. By Increasing coefficient K and f_r a closer and faster convergence to the desired distance of 10m can be achieved, albeit at the cost of increased fuel consumption. The Simulation preformed with the following coefficients:

k	5
k_r	500
f_r	3
α	0.3
d	4m

Table 4: Updated control coefficients and parameters for the second control approach

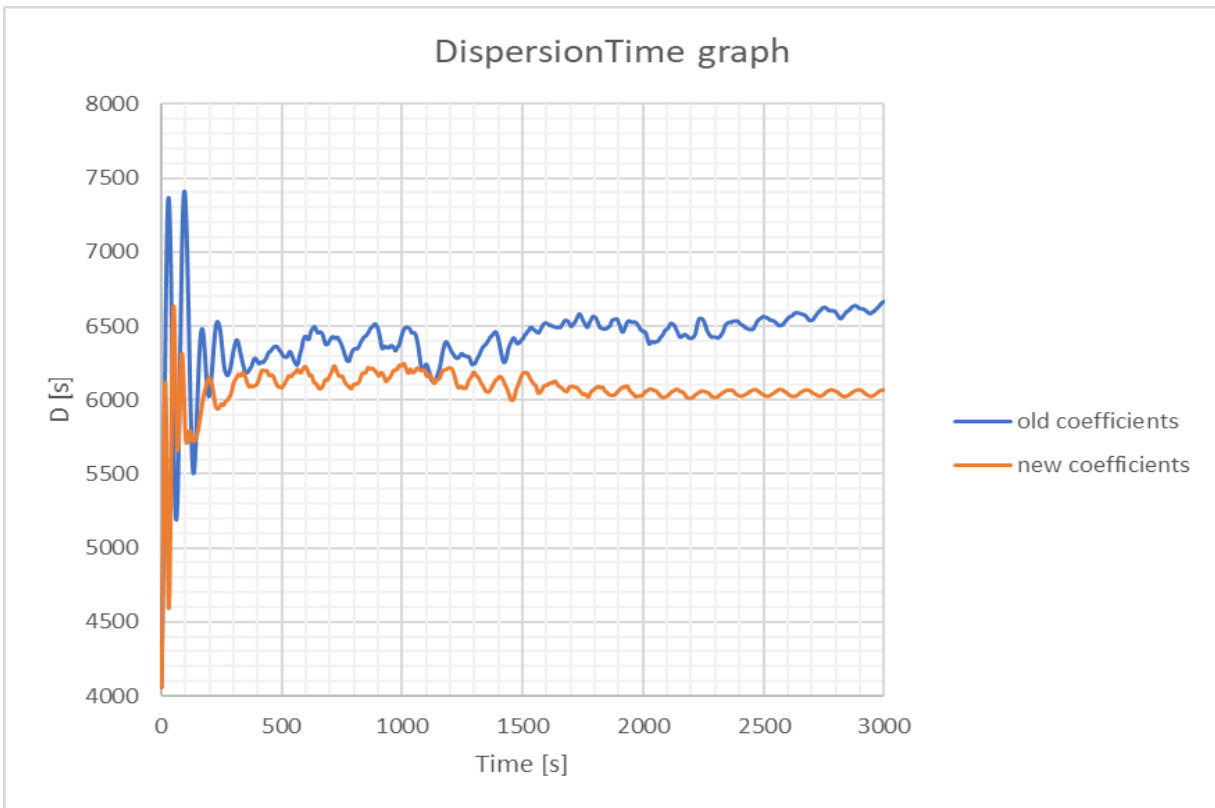


Figure 25: Fuel consumption time graph for a system of 20 satellites with good and bad coefficients

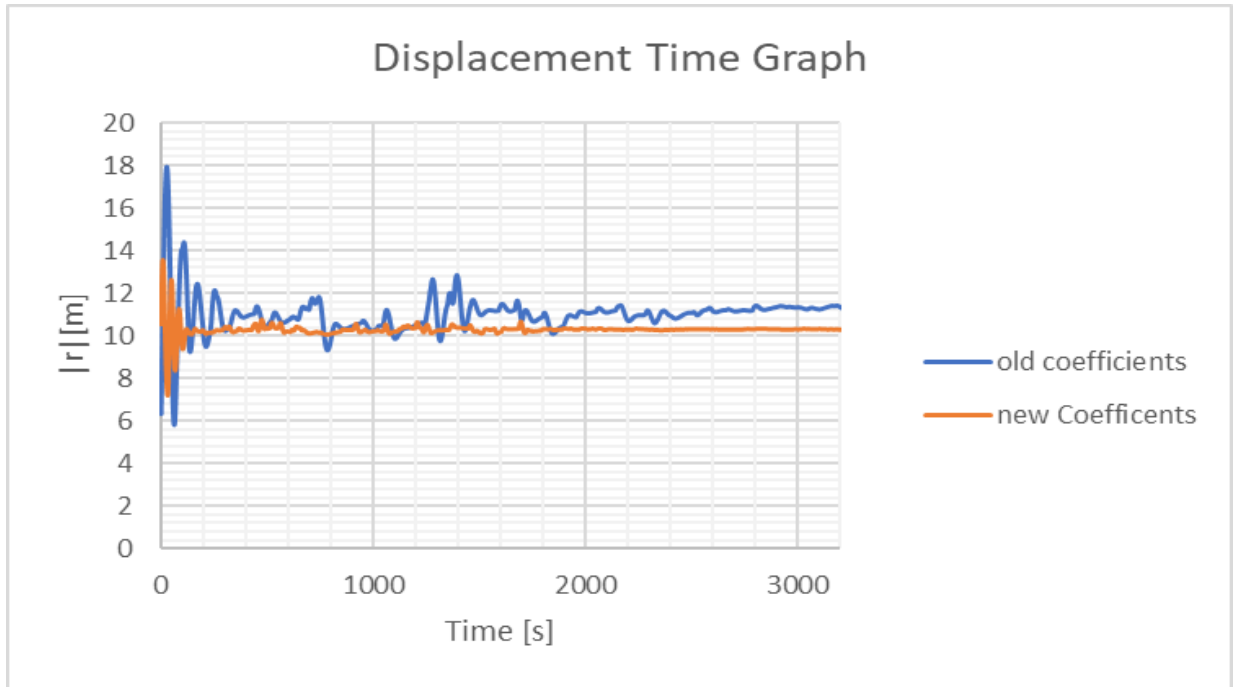


Figure 26: Displacement time graph on satellites with good and bad coefficients

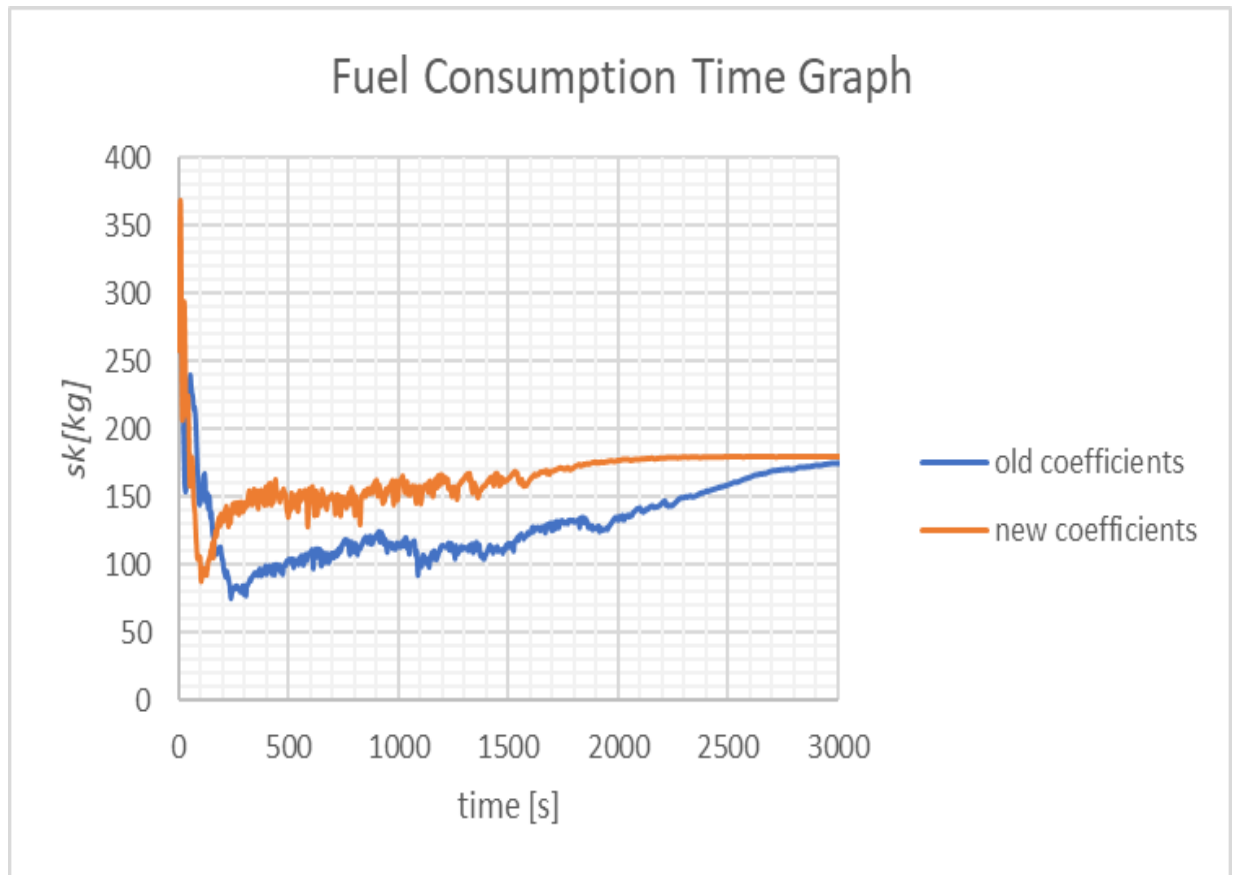


Figure 27: Fuel consumption time graph for a system of 20 satellites with good and bad coefficients

With the new coefficients, the displacement time graph, shown in Figure [26], is observed to converge to a value of approximately 10.3 meters, which is close to the initially predefined radius of the sphere formation of 10 meters. As previously noted, achieving more accurate values results in increased fuel consumption, as seen in Figure [27]. Due to the satellites in the system having a less convergent value, this results in a lower dispersion value, as indicated in Figure [25]. Overall, the Simulation with coefficients: $K = 5$, $k_r = 500$, $f_r = 3$, $\alpha = 0.3$, $d = 4\text{m}$, uses more fuel when compared with simulations with coefficients: $K = 1$, $k_r = 500$, $f_r = 1$, $\alpha = 0.3$, $d = 4\text{m}$. Depending on the situation, the distance a satellite should maintain from the sphere-formation center can be approximated by updating the coefficients. The final formation shape, with coefficients from table [4], is shown in figure [28].

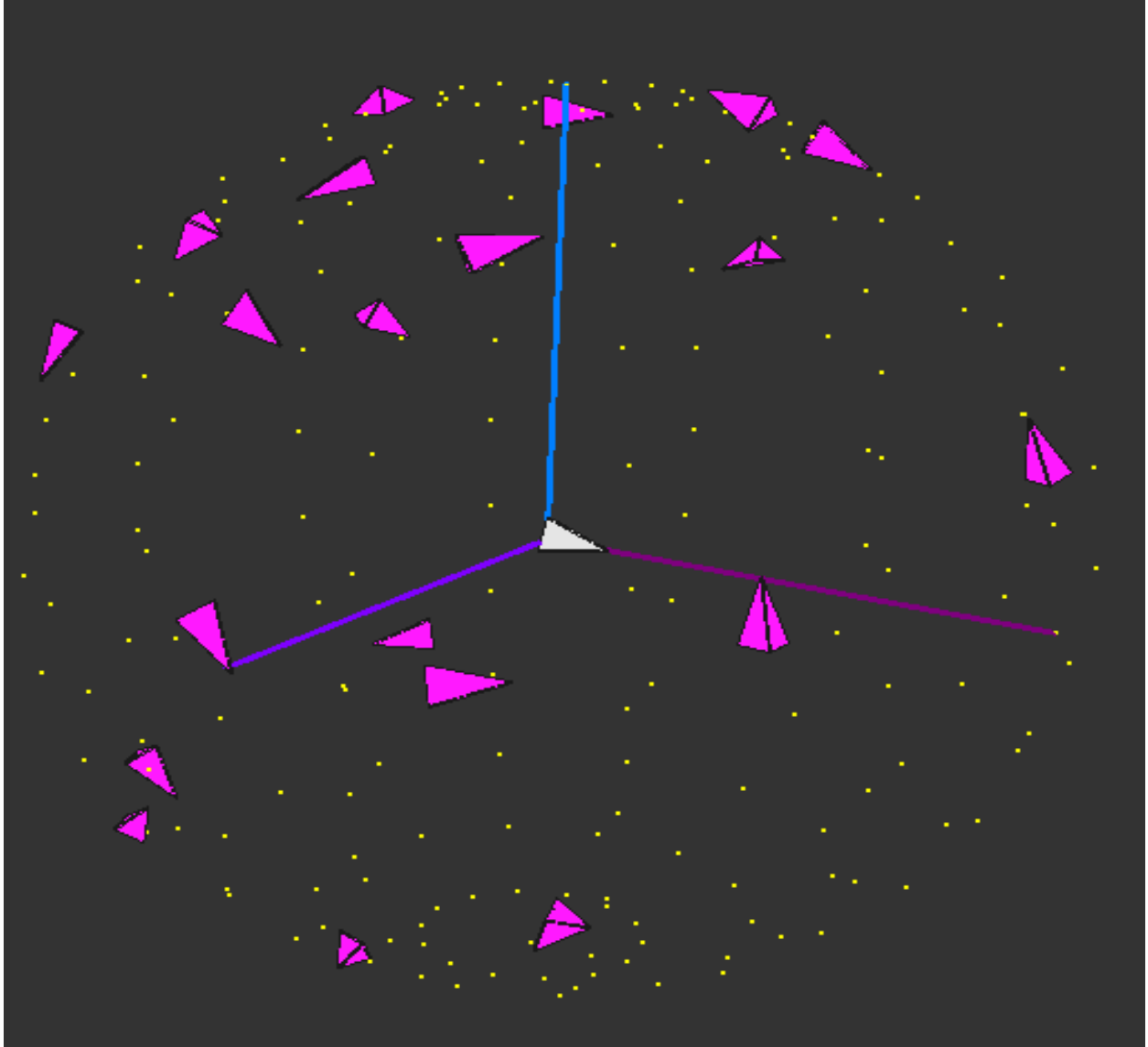


Figure 28: Model System of 20 satellites positioned on a sphere of radius 10 after 6000s simulation time, obtained after second control approach

The viability of this control has been established with the implementation of the Second Approach algorithm. The aim here is to control the model satellites to lie on a plane. The model satellites, as shown in the previous simulation (Figure [28]), already lie on the surface of a sphere. The sphere is a manifold, and the projection

onto a 2D plane is required. Using stereographic projection, which is a method of mapping points from a sphere onto a plane by extending a line from a point on the sphere (usually the North Pole) to the plane that is tangent to the sphere at the South Pole, the projection can be performed from the North Pole onto the plane $y = 0$. The model satellites have coordinates (x, y, z) on the sphere of radius R , and (X, Y) on the plane. The projection formula is provided as:

$$(X, Y) = \left(\frac{Rx}{R-z}, \frac{Ry}{R-z} \right)$$

(1.51)

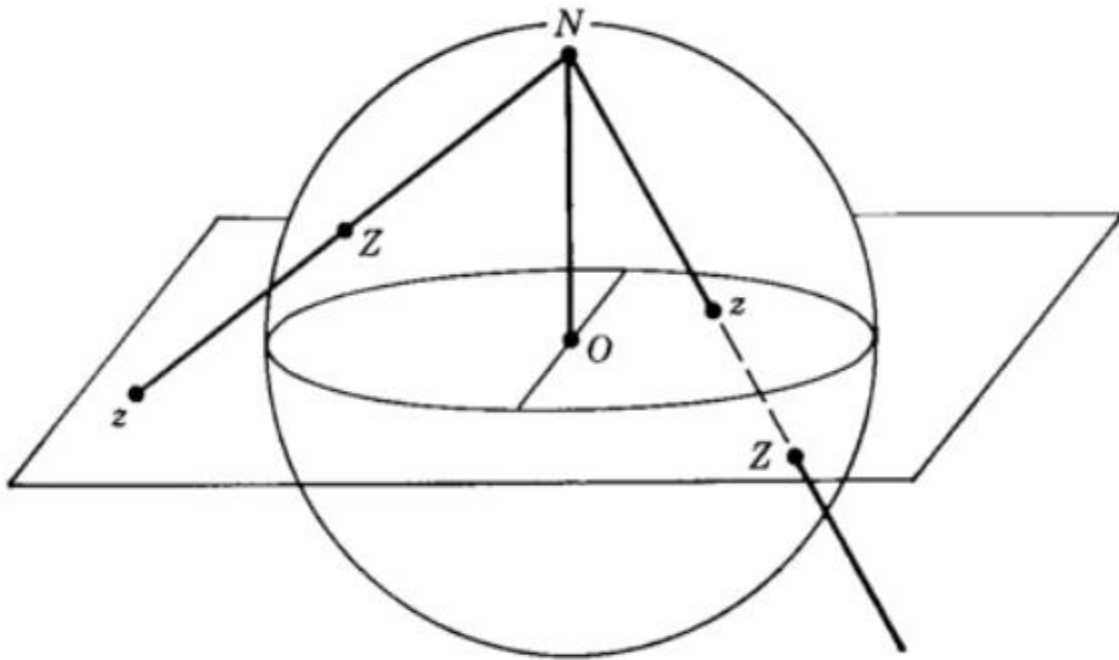


Figure 29: 3D illustration of a stereographic projection from the north pole onto a plane on the equator of a sphere [2]

Figure [29], depicts the projection of a point on the northpole of a sphere to a plane $y = 0$. After projection, from the sphere of radius 10m on to the plane $y = 0$, Figure [30] shows the projection of satellites with y coordinates $y < 0$ while and Figure [31] shows projection of satellites with y coordinates $y > 0$.

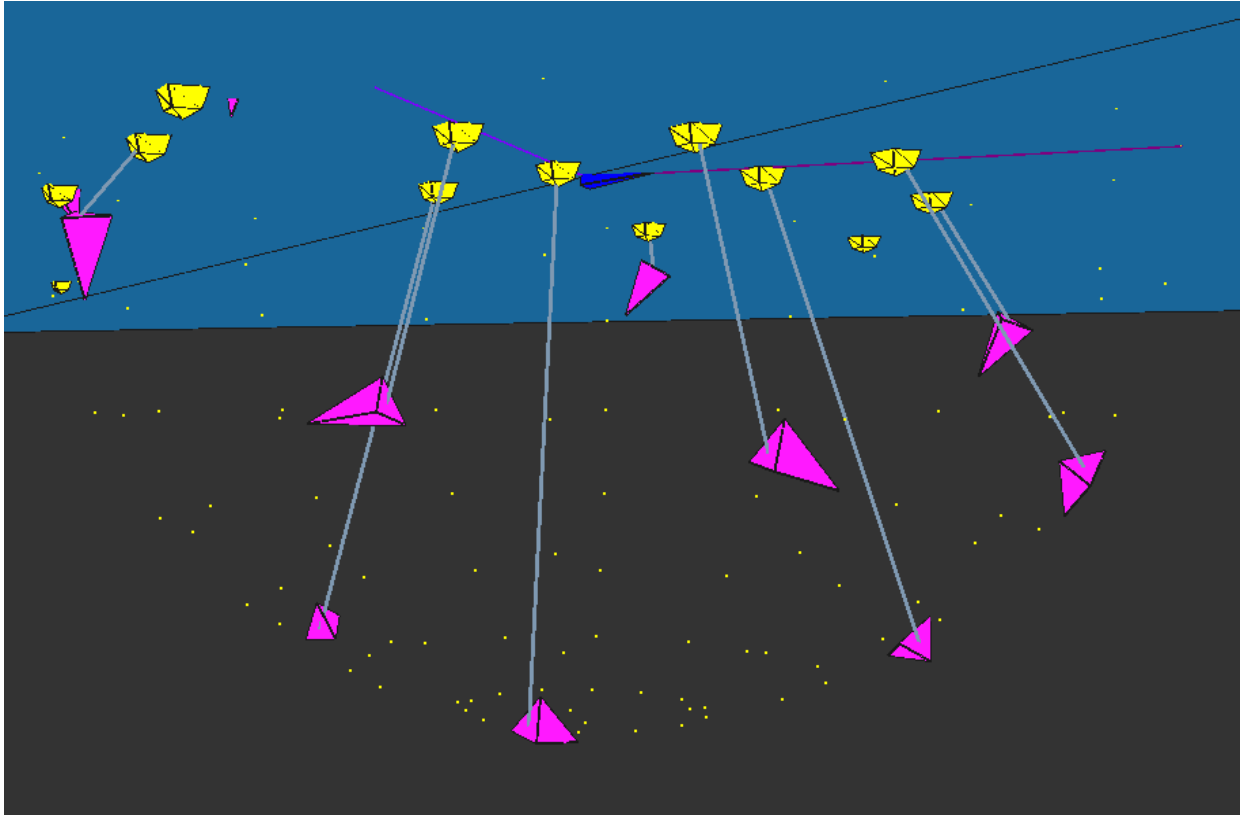


Figure 30: Projection of model satellites on the sphere of radius 10m and y coordinate < 0 on the plane $y = 0$ indicated by the blue plane. Projection points are indicated by the yellow sphere.

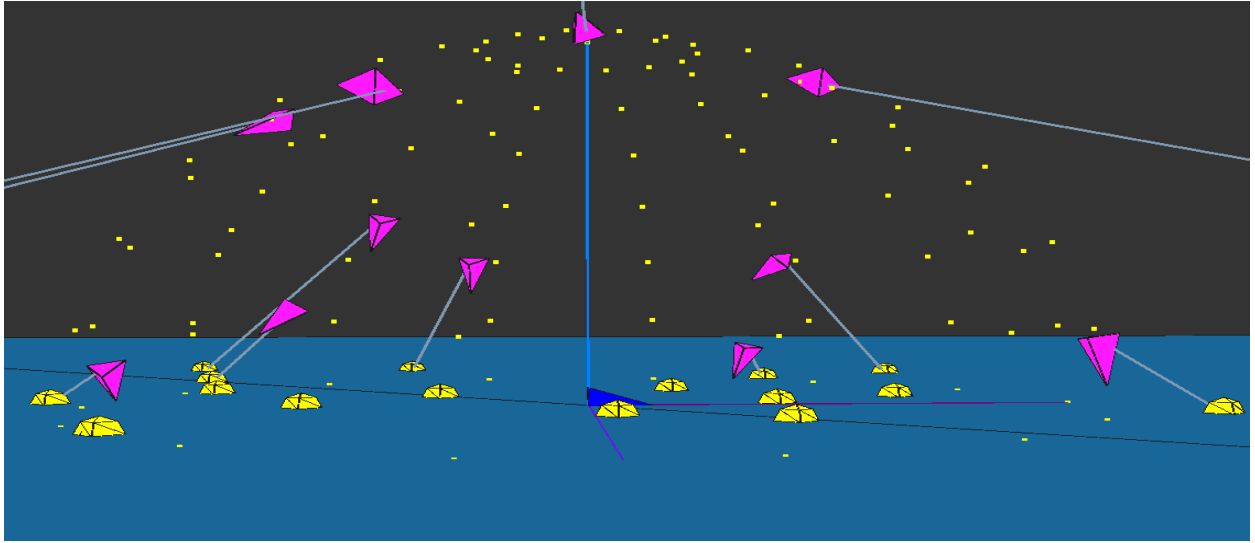


Figure 31: Projection of model satellites on the sphere of radius 10m and y coordinate > 0 on the plane $y=0$,

indicated by the blue plane. Projection points are indicated by the yellow sphere.

Using the control in equation (1.48) - (1.50), each satellite is controlled with its projection point as its goal location. The simulation with the coefficients,

k	0.5
k_r	500
f_r	1

Table 5: Control coefficients and parameters for the second control approach, controlling model satellites on a plane.

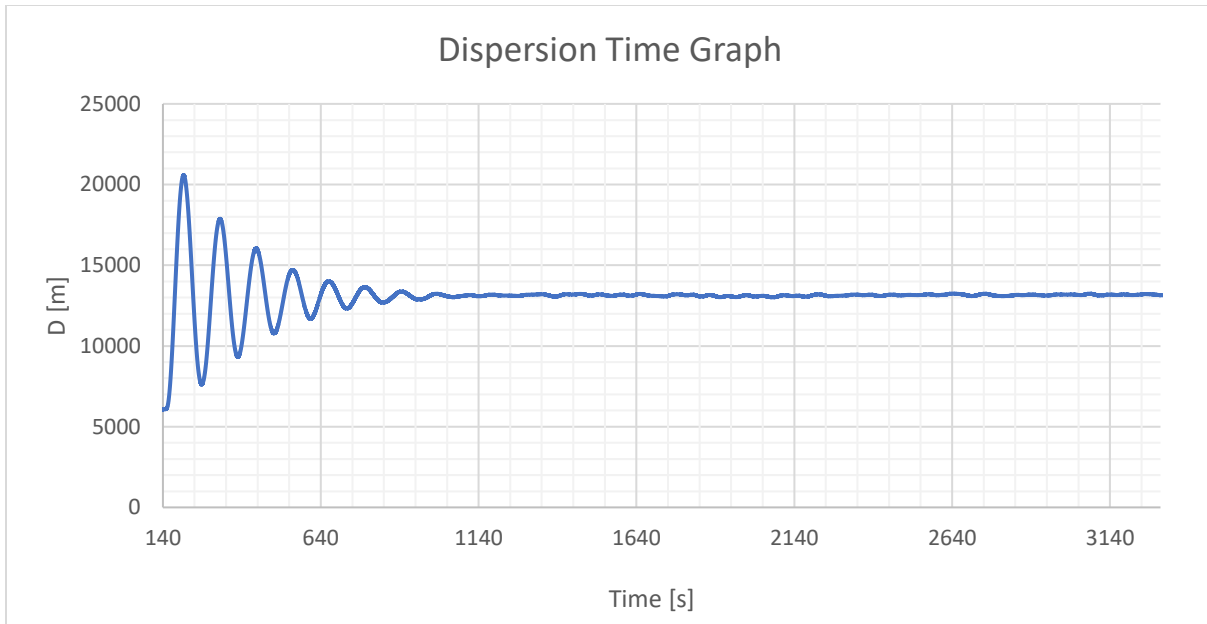


Figure 32: Dispersion time graph of model 20 satellites after control with projected points.

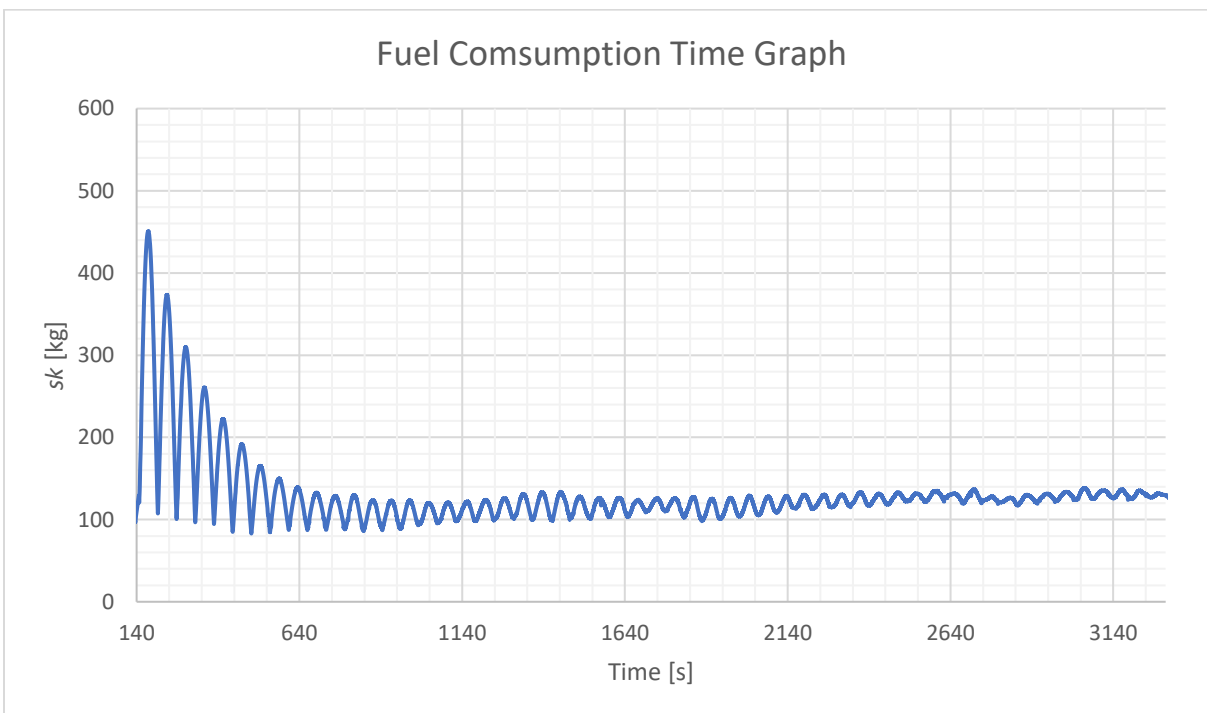


Figure 33: Fuel consumption time graph of model 20 satellites after control with projected points.

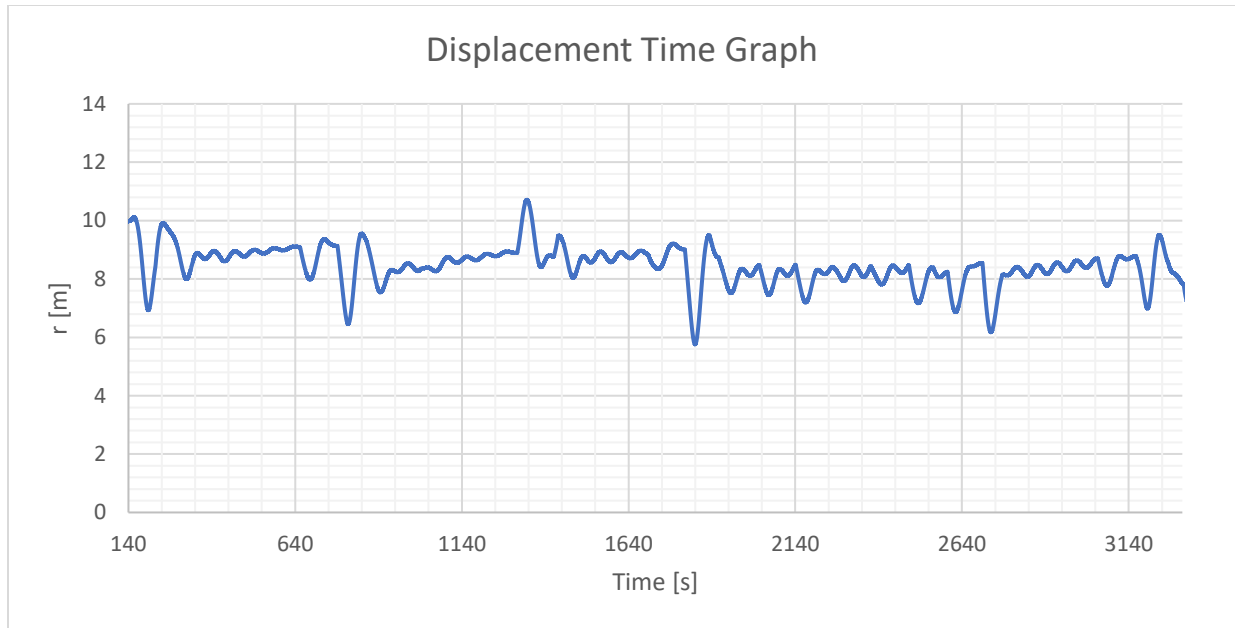


Figure 34: Displacement time graph of 1 model satellites from system, of 20 model satellites after control with projected points.

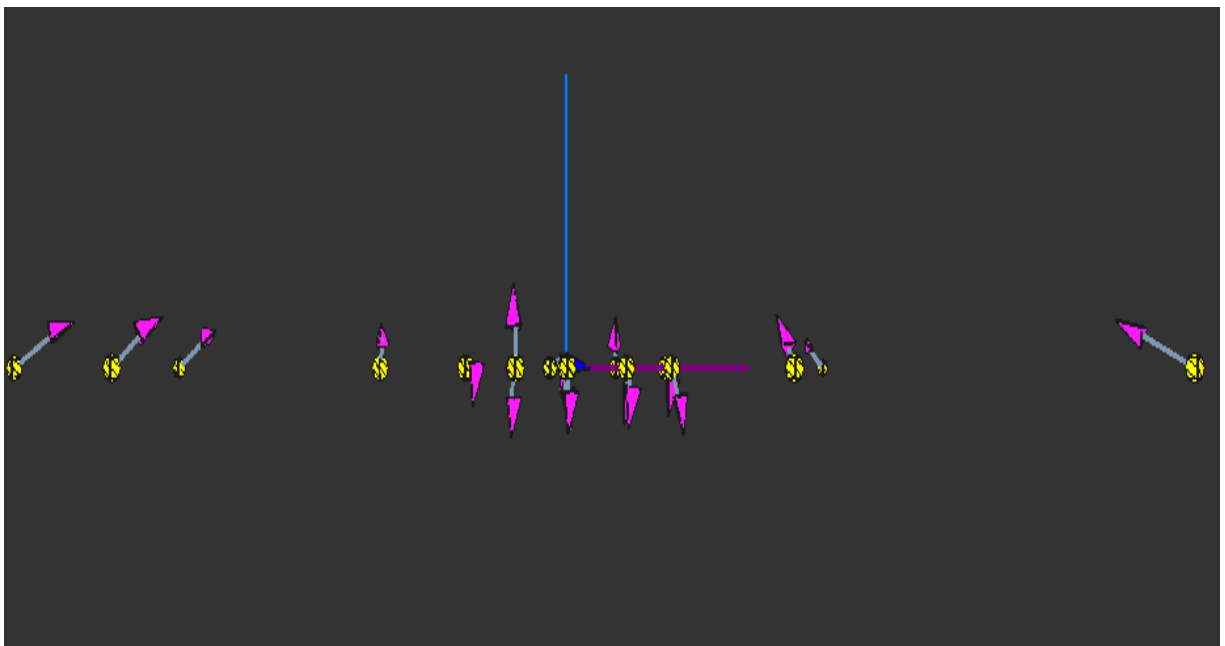


Figure 35: 2D view of the final formation of 20 model satellites after control with projected points on the plane $y=0$.

In Figure [32], the dispersion value is bounded, indicating that the control is effectively keeping the satellites on the plane $y=0$. In Figure [33], the oscillations in the curve are due to the satellites oscillating around their projected points. Increasing the value of k reduces these oscillations but increases fuel consumption. Figure [35] shows the final orientation of the satellite system on the plane $y=0$. As with the sphere, by increasing the value of k and f_r , the model satellites can be made to move faster and closer to their projected points, albeit at the cost of more fuel consumption. From Figure [34], the spikes in the displacement time graph are from the model avoiding obstacles, as it accelerates to correct its path.

4. Conclusion

This study investigated the maintenance of a model satellite swarm using decentralized control algorithms based on artificial potentials. The goal was to ensure viability, prevent divergence, and achieve stable formations. Two approaches were explored: the first maintains a stable formation through attractive and repulsive potentials, while the second fine-tunes these potentials for predefined formations. Evaluations include numerical analysis and real-time simulations using a custom 3D environment developed in C++, OpenGL, GLFW and GLEW.

- Two control algorithms approached to maintain space swarm for a model satellite swarm system control was proposed in the work, and their viability was considered.
- The first control approach results in convergence, but due the harmonic nature of the control, model satellites tend to oscillate. This oscillation results in unnecessary movement and increased fuel consumption.
- Despite the oscillations, this approach is useful for a system of satellites with obstacles in its path. It allows the satellites to correct and update its formation based on its surrounding.
- The control coefficients highly depend on the systems initial conditions (position and velocity). Poor initial coefficients could lead to a large dispersion value, requiring increased fuel consumption to correct its motion.

- The second control approach results in convergence. It is characterized by a more stable convergence compared to the first approach. This method avoids unnecessary movement, leading to more efficient fuel consumption.
- This approach allows predetermination of formation shapes, demonstrated by controlling the system to form a sphere and later project onto a plane. This capability is crucial for various satellite missions where specific formations are required.

This study is significant for developing decentralized control algorithms for satellite swarms, this is crucial for Earth observation, environmental monitoring, debris removal. Decentralized methods offer several advantages over centralized overcome the limitations of centralized control, such as communication delays and potential failures, and meet the autonomy requirements of satellites. Artificial potential fields help ensure cohesive formation and effective navigation. Future research can refine these algorithms, explore more formation shapes, and optimize control coefficients for better fuel efficiency and stability. Future research can build upon the findings of this study by:

- Finding methods to optimize control coefficients which can help achieve better fuel efficiency and stability, reducing the operational costs and extending the mission lifespan of satellite swarms.

- Finding and implemented different non harmonic potentials, which should eliminate some of the oscillations in the motion of the satellites.
- More complex algorithms could potentially yield even better results, offering further stability and control over satellite formations

5. References

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